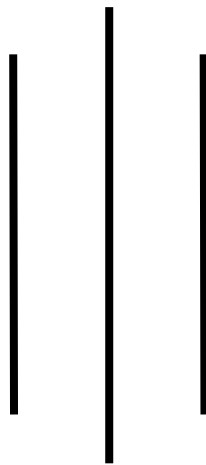


TRIBHUVAN UNIVERSITY

PATAN MULTIPLE CAMPUS

PATANDHOKA, LALITPUR



DATABASE ADMINISTRATION (BIT 352)

SUBMITTED BY:

NAME: SURESH DAHAL

ROLL: 23

CLASS: BIT-III/II

DATE: 2083/04/06

SUBMITTED TO:

BIPIN TIMALSINA

CHECKED BY

1. EXPLORING THE MULTITENANT ARCHITECTURE OF ORACLE 19C.

1. Display current container

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> show con_name;

CON_NAME
-----
CDB$ROOT
```

2. Display information about the Container Database

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> select name, cdb, open_mode from v$database;

NAME          CDB OPEN_MODE
-----
ORCL          YES READ WRITE
```

3. Display all Pluggable Databases

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> show pdbs;

  CON_ID CON_NAME          OPEN MODE RESTRICTED
-----
      2 PDB$SEED             READ ONLY NO
      3 ORCLPDB            READ WRITE NO
```

4. Switch from the Root Container to a Pluggable Database

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> alter session set container = orclpdb;

Session altered.
```

5. Verify the current container

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> show con_name;

CON_NAME
-----
ORCLPDB
```

6. Return to the Root Container

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> alter session set container = cdb$root;

Session altered.

SQL> show con_name;

CON_NAME
-----
CDB$ROOT
```

7. View the open mode of all PDBs

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> show pdbs;

  CON_ID CON_NAME              OPEN MODE  RESTRICTED
  -----
        2 PDB$SEED              READ ONLY  NO
        3 ORCLPDB               READ WRITE NO
```

8. Change the open mode of a PDB

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> alter pluggable database orclpdb close immediate;

Pluggable database altered.

SQL> show pdbs;

  CON_ID CON_NAME              OPEN MODE  RESTRICTED
  -----
        2 PDB$SEED              READ ONLY  NO
        3 ORCLPDB               MOUNTED
SQL> alter pluggable database orclpdb open;

Pluggable database altered.

SQL> show pdbs;

  CON_ID CON_NAME              OPEN MODE  RESTRICTED
  -----
        2 PDB$SEED              READ ONLY  NO
        3 ORCLPDB               READ WRITE NO
SQL>
```

2. WORKING WITH STARTUP AND SHUTDOWN MODES IN ORACLE.

STARTUP MODES:

1. Startup database normally

```
SQL> startup;
ORACLE instance started.

Total System Global Area 4915720712 bytes
Fixed Size                  9277960 bytes
Variable Size               989855744 bytes
Database Buffers            3909091328 bytes
Redo Buffers                 7495680 bytes
Database mounted.
Database opened.
```

2. Startup database in NOMOUNT mode

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> shutdown immediate;
Database closed.
Database dismounted.
ORACLE instance shut down.
SQL> startup nomount;
ORACLE instance started.

Total System Global Area 4915720712 bytes
Fixed Size                  9277960 bytes
Variable Size               989855744 bytes
Database Buffers            3909091328 bytes
Redo Buffers                 7495680 bytes
```

3. Startup database in MOUNT mode

```
SQL> startup mount;
ORACLE instance started.

Total System Global Area 4915720712 bytes
Fixed Size                  9277960 bytes
Variable Size               989855744 bytes
Database Buffers            3909091328 bytes
Redo Buffers                 7495680 bytes
Database mounted.
```

4. Open the database

```
SQL> alter database open;

Database altered.

SQL> set serveroutput on;
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A
```

5. Startup force

```
SQL> startup force;
ORACLE instance started.

Total System Global Area 4915720712 bytes
Fixed Size                  9277960 bytes
Variable Size               989855744 bytes
Database Buffers           3909091328 bytes
Redo Buffers                 7495680 bytes
Database mounted.
Database opened.
SQL> exec my_info;

PL/SQL procedure successfully completed.

SQL> set serveroutput on;
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A
```

6. Suspend database

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> alter system suspend;

System altered.
```

7. Resume database

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> alter system resume;

System altered.
```

SHUTDOWN MODES:

1. Shutdown normal

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> shutdown normal;
Database closed.
Database dismounted.
ORACLE instance shut down.
```

2. Shutdown transactional

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> shutdown transactional
Database closed.
Database dismounted.
ORACLE instance shut down.
```

3. Shutdown immediate

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> shutdown immediate
Database closed.
Database dismounted.
ORACLE instance shut down.
```

4. Shutdown abort

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

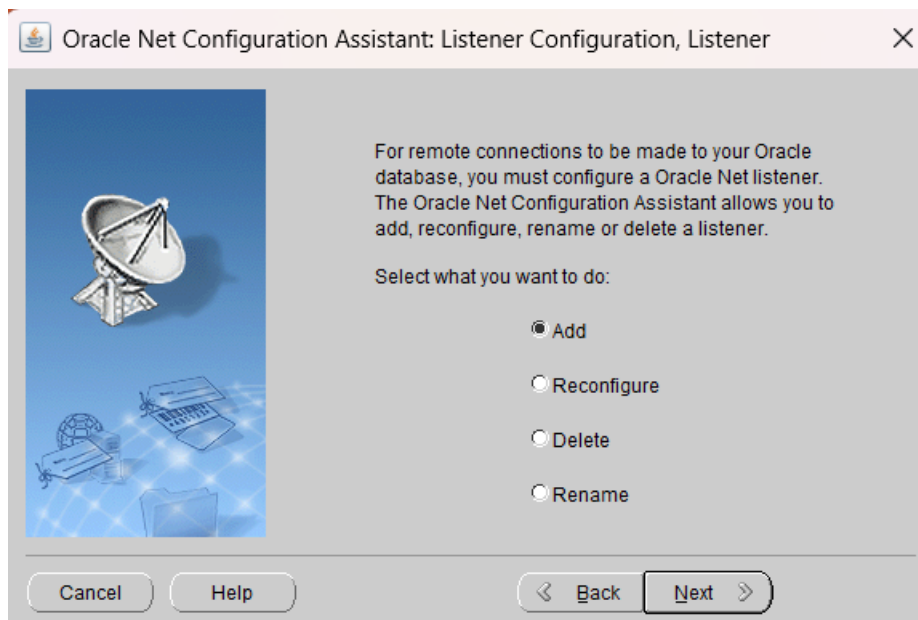
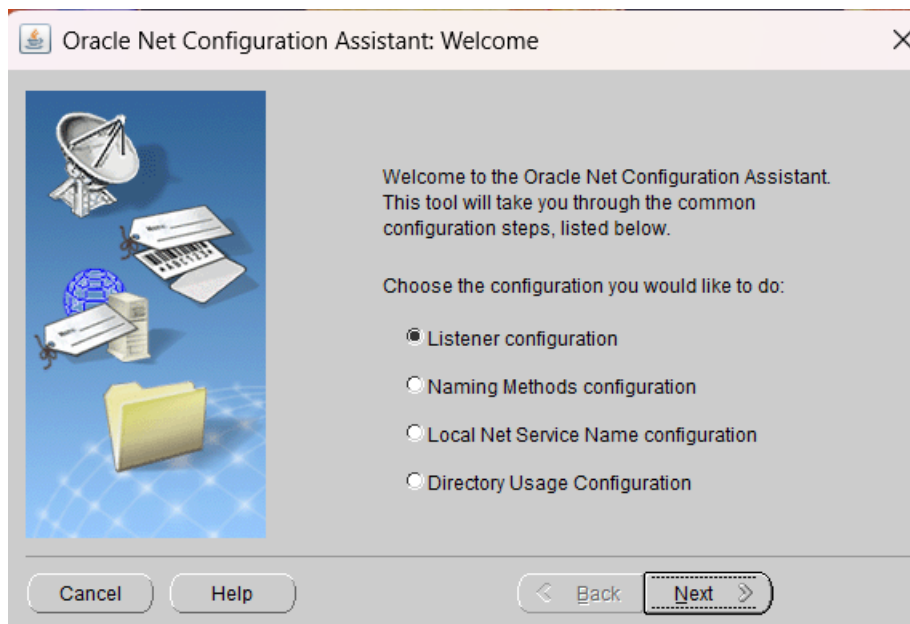
PL/SQL procedure successfully completed.

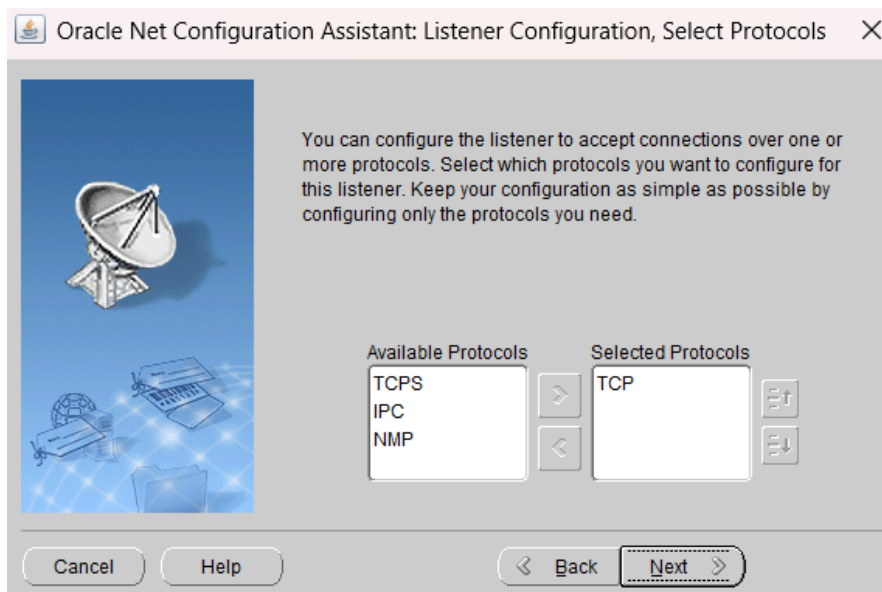
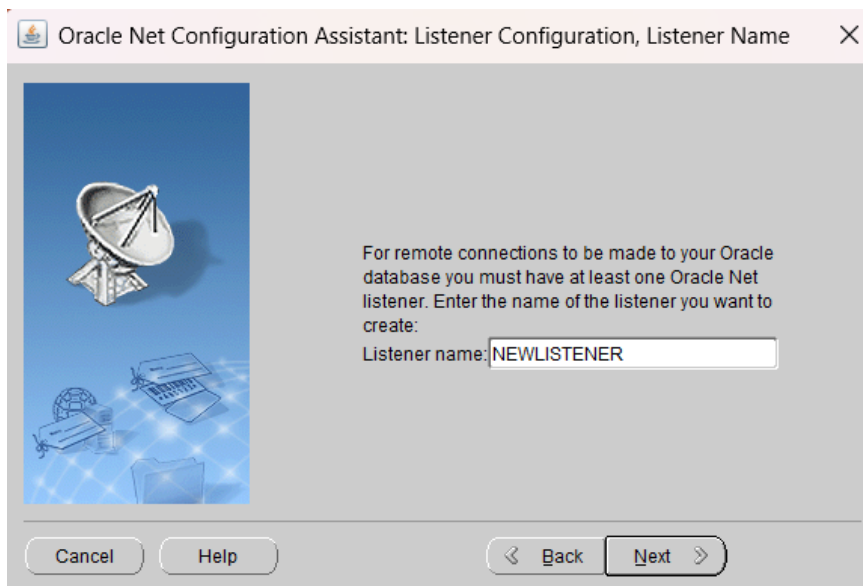
SQL> shutdown abort
ORACLE instance shut down.
```

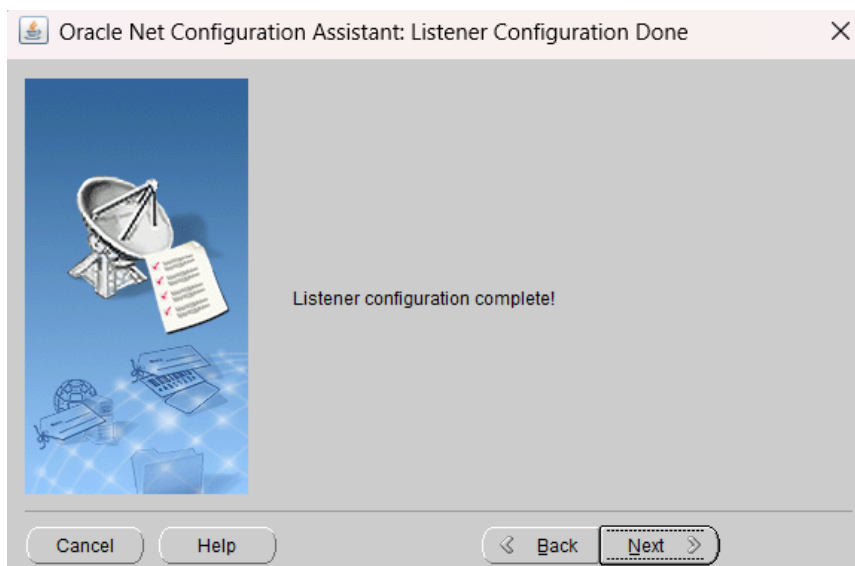
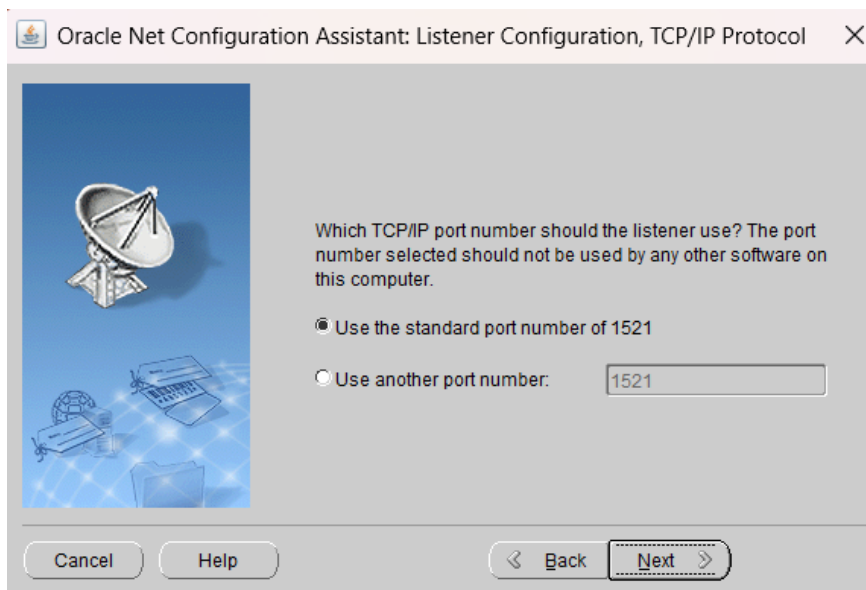
3. USING ORACLE NET CONFIGURATION ASSISTANT (NETCA) FOR NET SERVICE CONFIGURATION.

PART A: Listener Configuration

Step 1: Add a Listener







Step 2: Test the listener

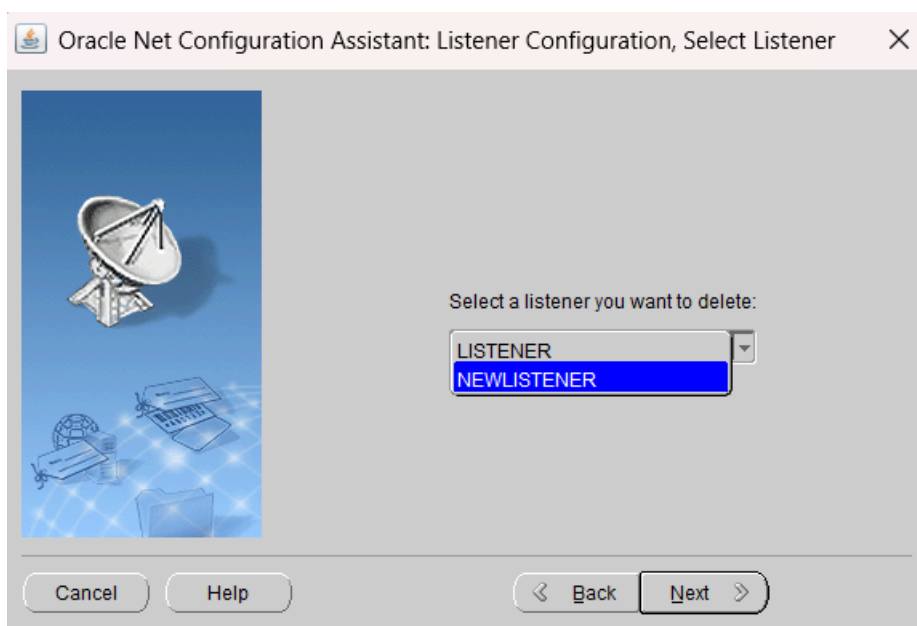
```
C:\Users\VICTUS>lsnrctl status NEWLISTENER

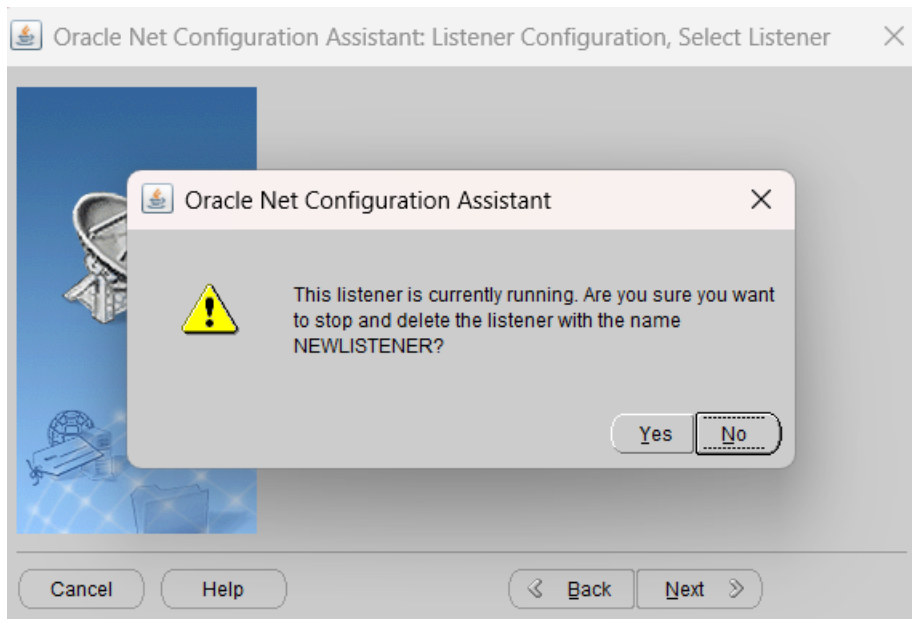
LSNRCTL for 64-bit Windows: Version 19.0.0.0.0 - Production on 12-JUL-2026 15:00:55

Copyright (c) 1991, 2019, Oracle. All rights reserved.

Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=192.168.1.118)(PORT=1521)))
STATUS of the LISTENER
-----
Alias                     LISTENER
Version                  TNSLSNR for 64-bit Windows: Version 19.0.0.0.0 - Production
Start Date               21-JUN-2026 08:45:07
Uptime                   21 days 6 hr. 15 min. 48 sec
Trace Level              off
Security                 ON: Local OS Authentication
SNMP                     OFF
Listener Parameter File   C:\Oracle_19C\network\admin\listener.ora
Listener Log File        C:\Oracle19C\diag\tnslnsr\Aconitin\listener>alert\log.xml
Listening Endpoints Summary...
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)(HOST=Aconitin)(PORT=1521)))
  (DESCRIPTION=(ADDRESS=(PROTOCOL=ipc)(PIPENAME=\\.\pipe\EXTPROC1521ipc)))
Services Summary...
Service "CLRExtProc" has 1 instance(s).
  Instance "CLRExtProc", status UNKNOWN, has 1 handler(s) for this service...
```

Step 3: Delete the listener





Verify whether the listener has been deleted

```
C:\Users\VICTUS\suresh>lsnrctl status NEWLISTENER

LSNRCTL for 64-bit Windows: Version 19.0.0.0.0 - Production on 12-JUL-2026 15:06:03

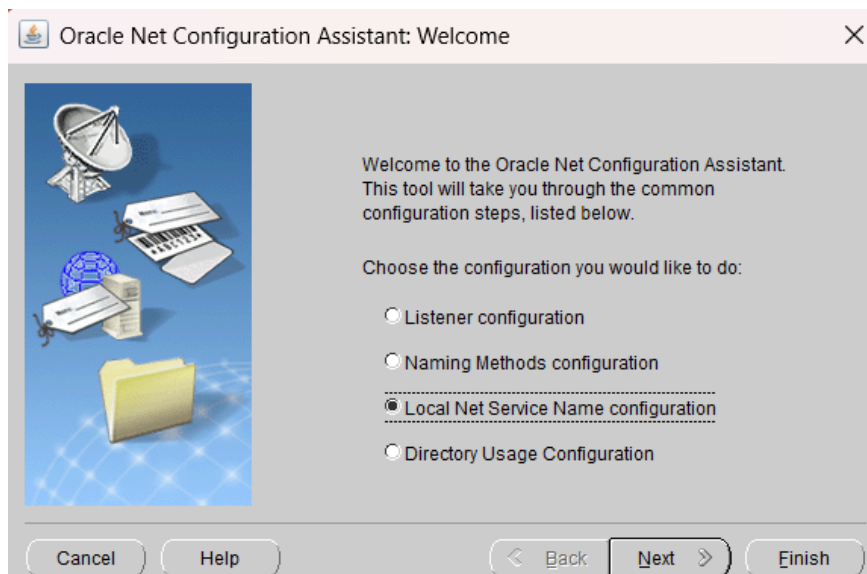
Copyright (c) 1991, 2019, Oracle. All rights reserved.

TNS-01101: Could not find listener name or service name NEWLISTENER
```

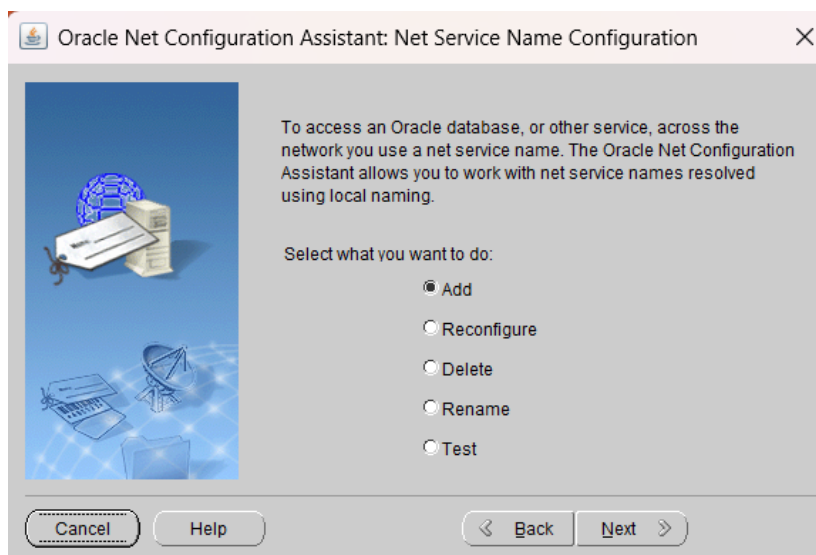
Part B: Local Net Service Name Configuration

Step 4: Add a Local Net Service Name

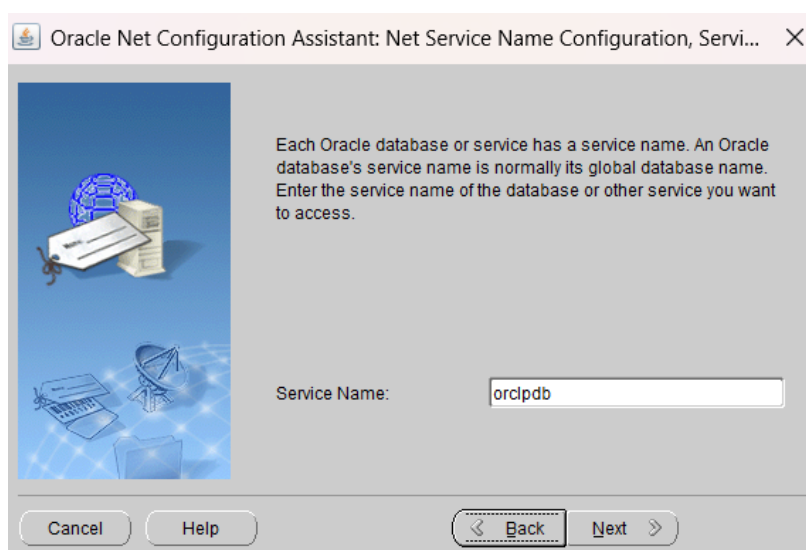
1. Select **Local Net Service Name Configuration**.



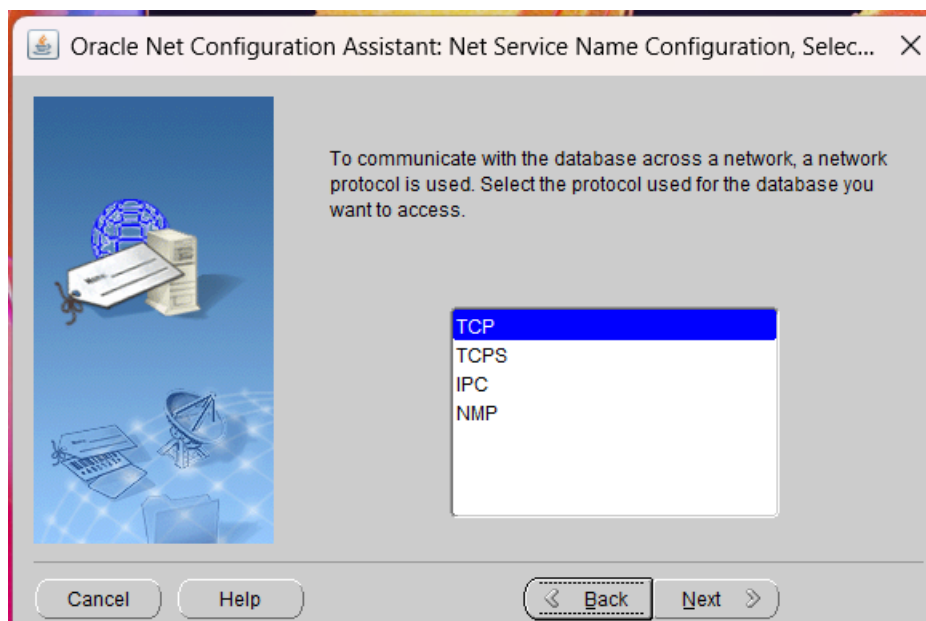
2. Click **Add**.



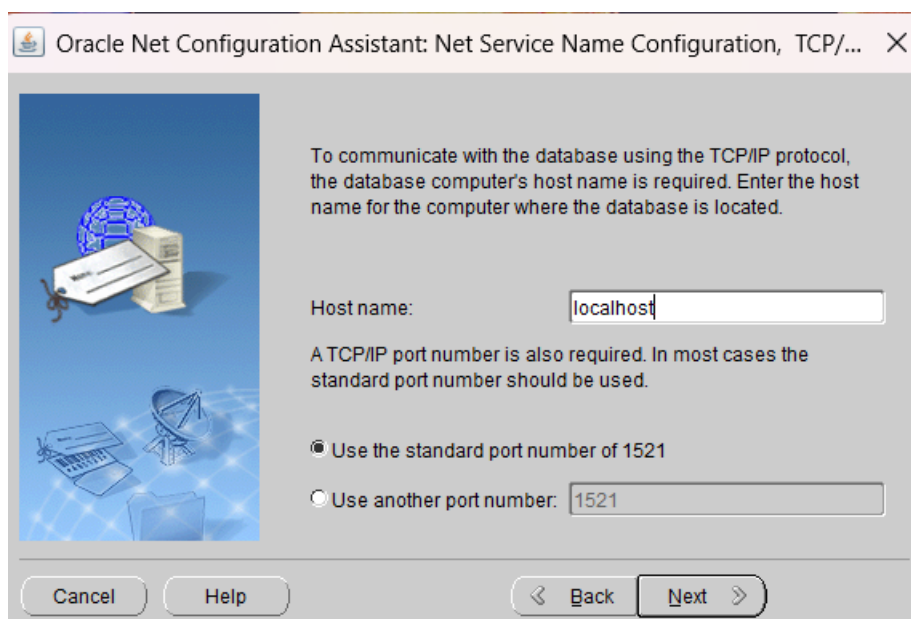
3. Enter the Service Name



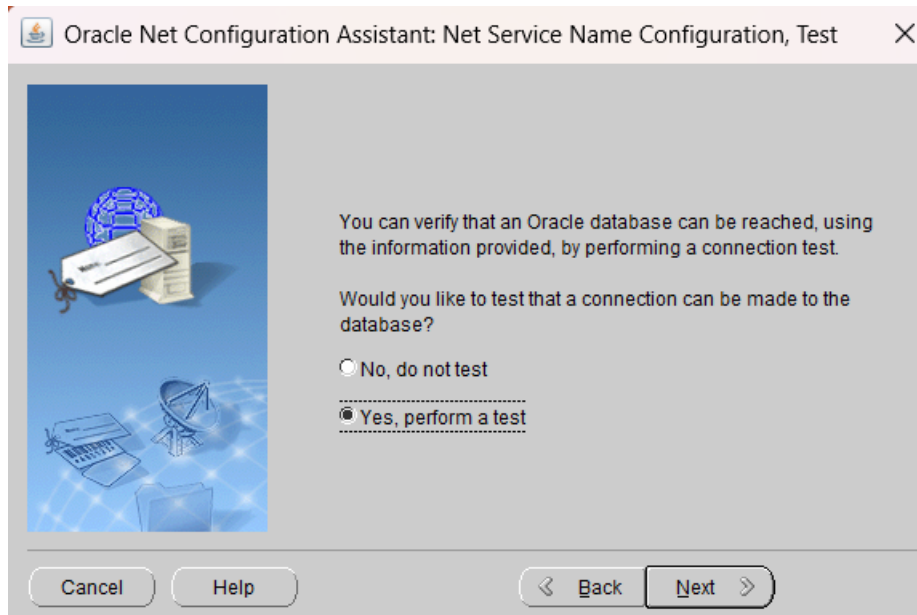
4. Select **TCP** as the protocol.



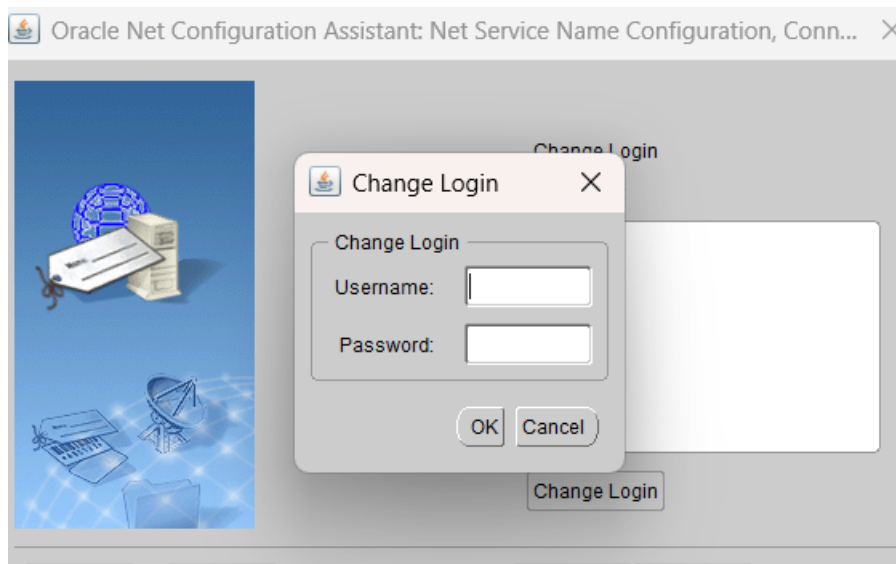
5. Enter the Host Name

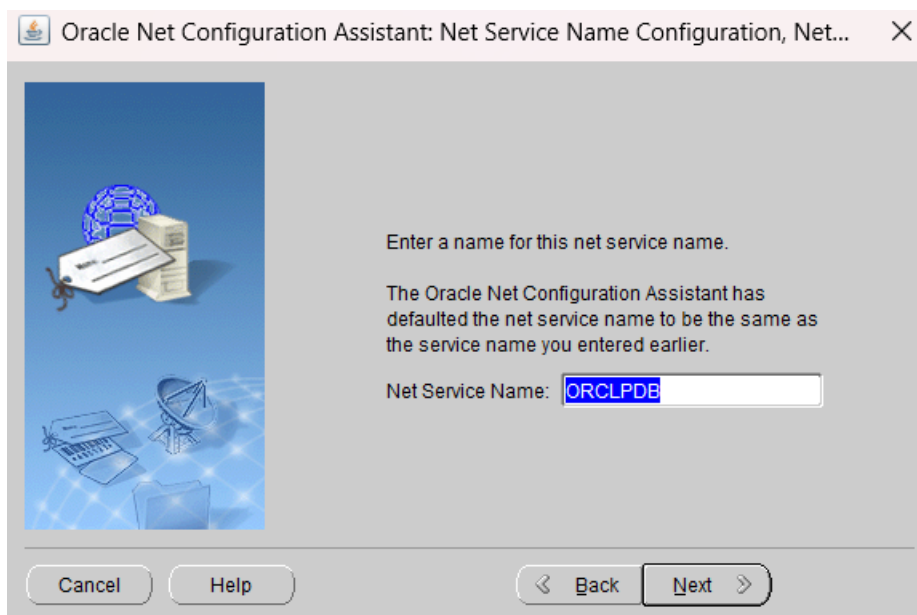
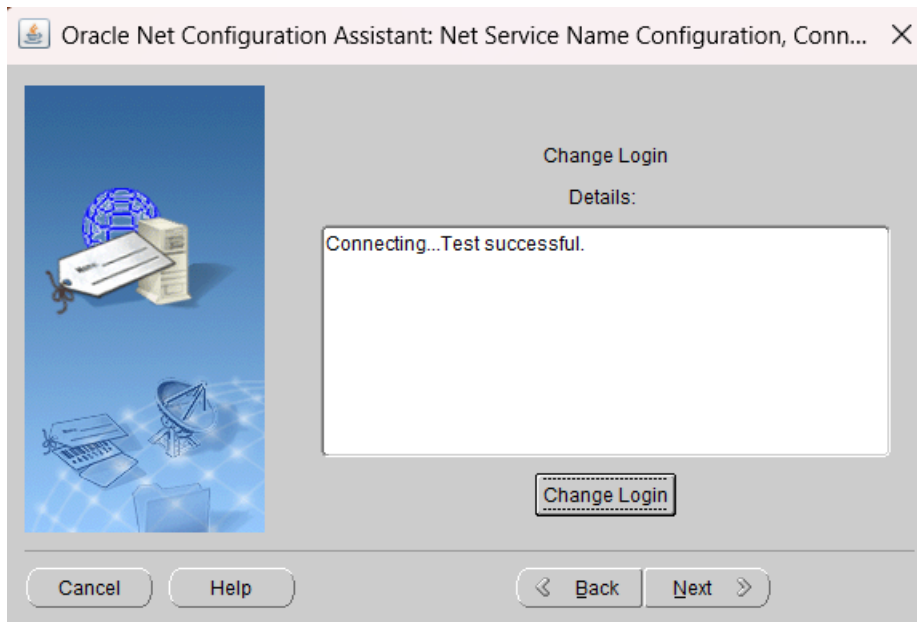


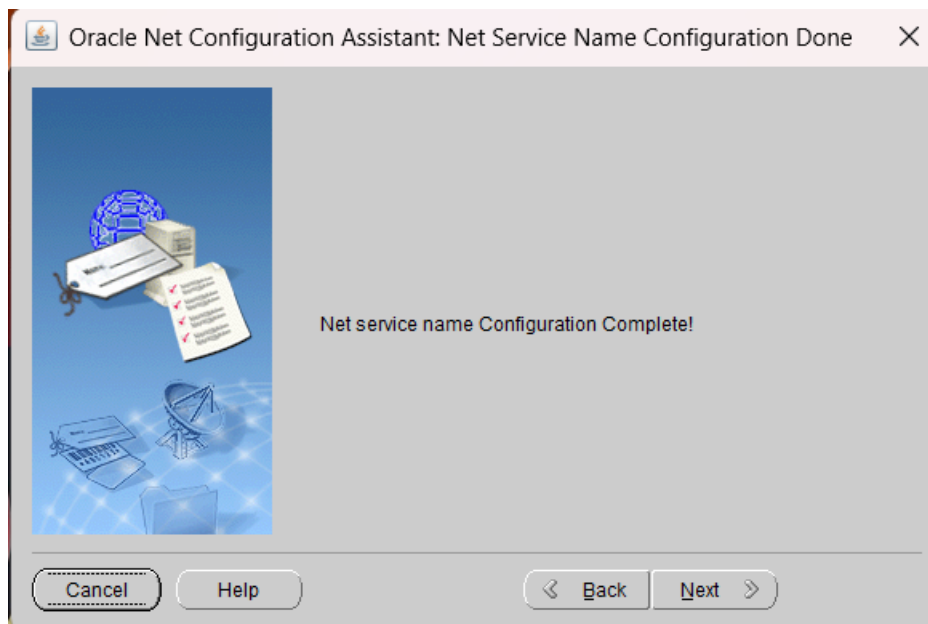
6. Test the connection.



7. Enter system username and password





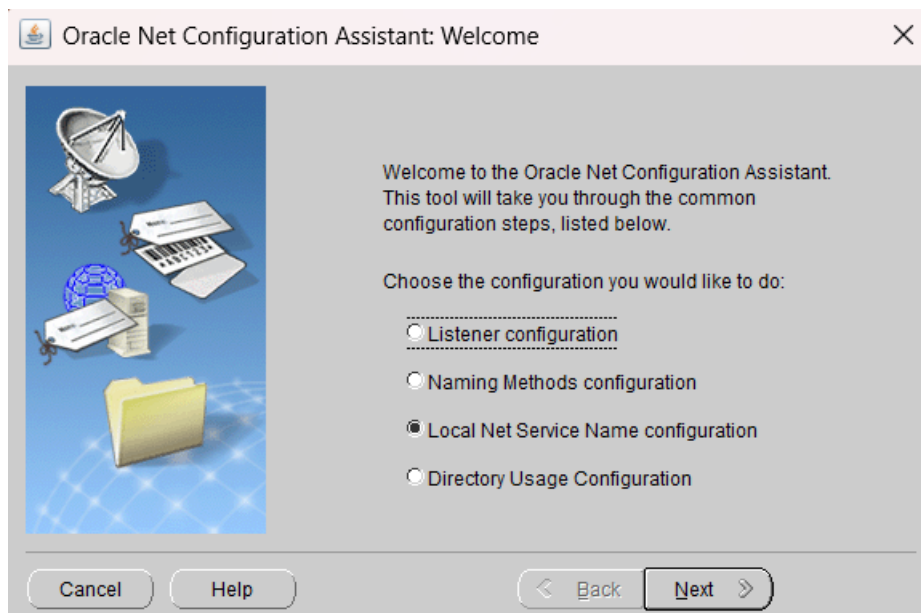


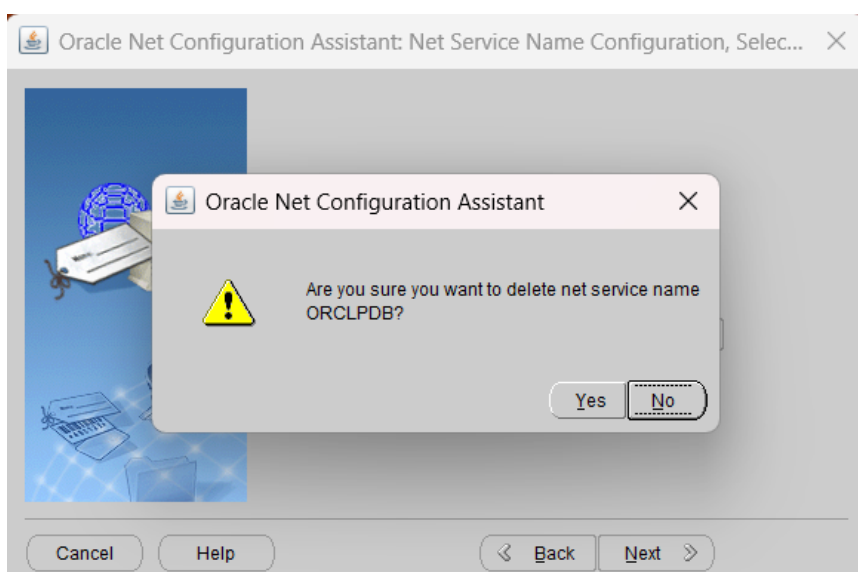
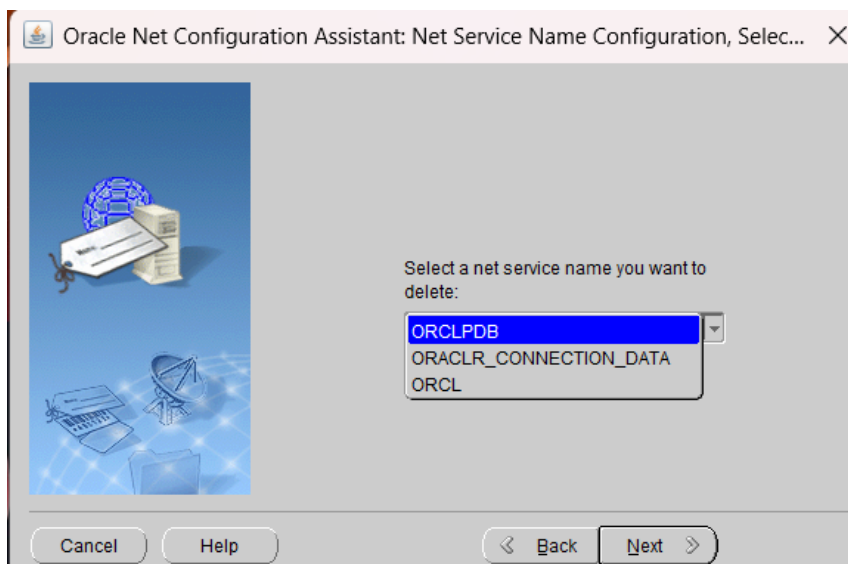
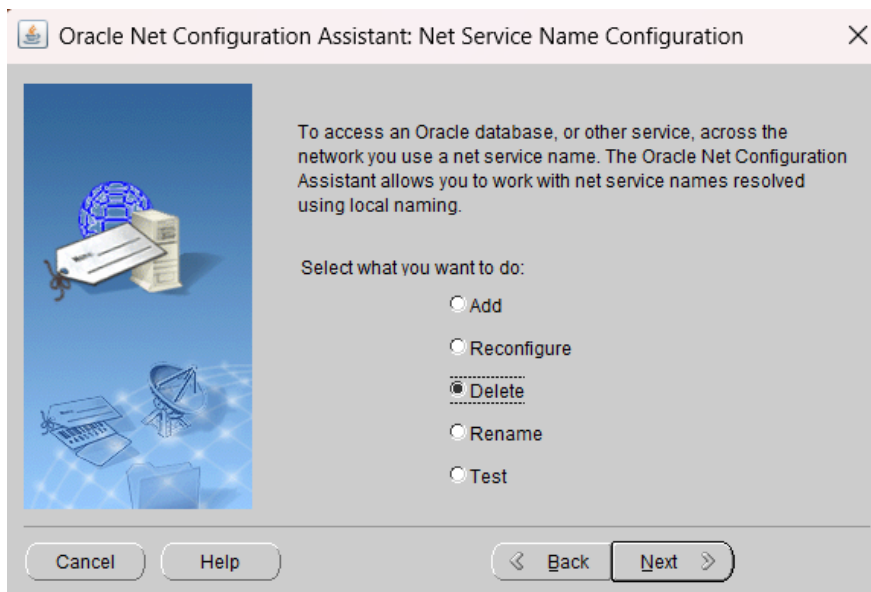
Step 5: Test the Local Net Service Name

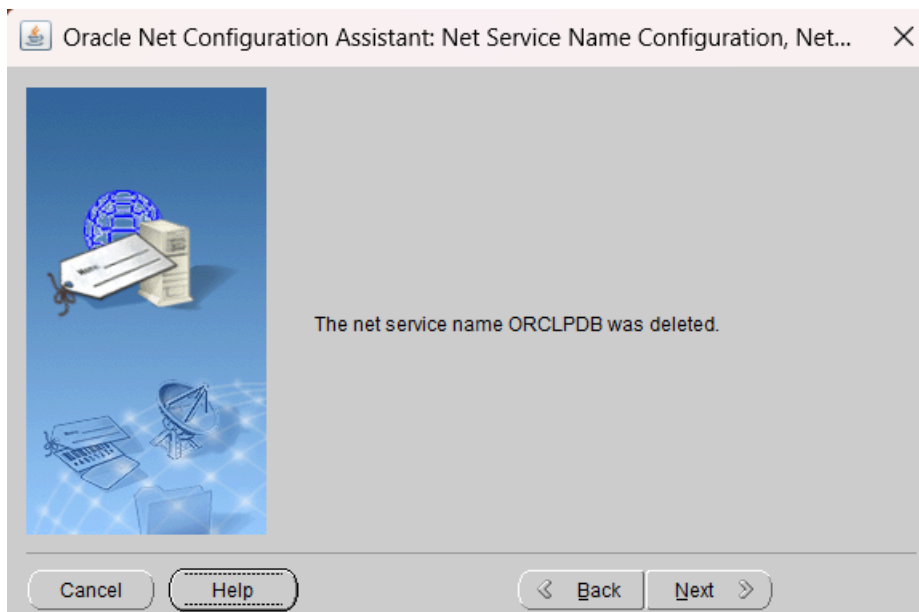
```
SQL> connect system/thaxaena@orclpdb;
Connected.
SQL> show con_name;

CON_NAME
-----
ORCLPDB
```

Step 6: Delete the Local Net Service Name







Verify delete

```
SQL> connect system/thaxaena@orclpdb;
ERROR:
ORA-12154: TNS:could not resolve the connect identifier specified

Warning: You are no longer connected to ORACLE.
```

Part C: Listener Control Commands

Step 7: Start the Listener

```
C:\Users\VICTUS\suresh>lsnrctl start

LSNRCTL for 64-bit Windows: Version 19.0.0.0.0 - Production on 12-JUL-2026 15:28:06

Copyright (c) 1991, 2019, Oracle. All rights reserved.

TNS-01106: Listener using listener name LISTENER has already been started
```

Step 8: Check Listener Status

```
C:\Users\VICTUS\suresh>lsnrctl status

LSNRCTL for 64-bit Windows: Version 19.0.0.0.0 - Production on 12-JUL-2026 15:28:28

Copyright (c) 1991, 2019, Oracle. All rights reserved.

Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=localhost)(PORT=1521)))
STATUS of the LISTENER
-----
Alias                     LISTENER
Version                   TNSLSNR for 64-bit Windows: Version 19.0.0.0.0 - Production
Start Date                21-JUN-2026 08:45:07
Uptime                    21 days 6 hr. 43 min. 21 sec
Trace Level               off
Security                  ON: Local OS Authentication
SNMP                      OFF
Listener Parameter File   C:\Oracle_19C\network\admin\listener.ora
Listener Log File         C:\Oracle19C\diag\tnslsnr\Aconitin\listener>alert\log.xml
```

Step 9: Reload the Listener

```
C:\Users\VICTUS\suresh>lsnrctl reload

LSNRCTL for 64-bit Windows: Version 19.0.0.0.0 - Production on 12-JUL-2026 15:30:44

Copyright (c) 1991, 2019, Oracle. All rights reserved.

Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=localhost)(PORT=1521)))
The command completed successfully
```

Step 10: Stop the Listener

```
C:\Users\VICTUS\suresh>lsnrctl stop

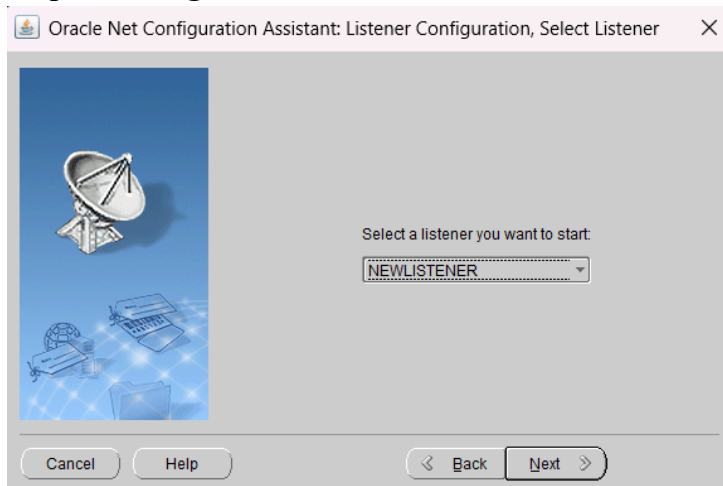
LSNRCTL for 64-bit Windows: Version 19.0.0.0.0 - Production on 12-JUL-2026 15:31:02

Copyright (c) 1991, 2019, Oracle. All rights reserved.

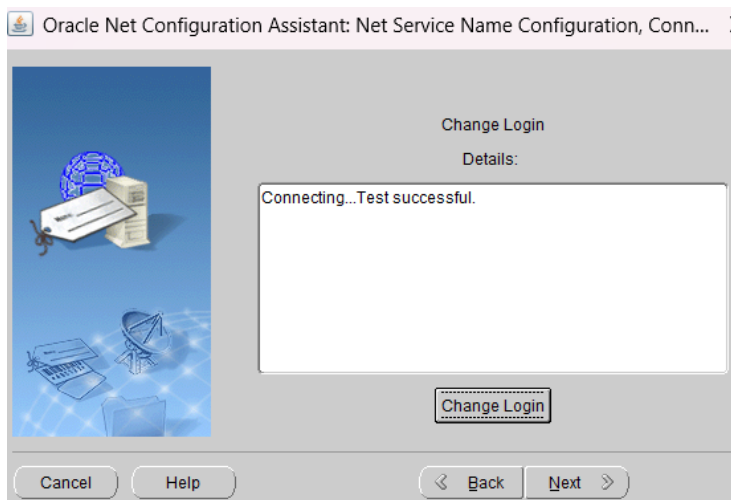
Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=localhost)(PORT=1521)))
The command completed successfully
```

4. UNLOCKING HR-SCHEMA IN ORACLE 19C AND CONNECTING WITH HR USER.

Step 1: Configure listener



Test the connection



Step 2: Connect as SYSDBA

```
C:\Users\VICTUS>sqlplus / as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Sun Jul 12 15:53:47 2026
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle. All rights reserved.

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0
```

Step 3: Verify the Current Container

```
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> show con_name;

CON_NAME
-----
CDB$ROOT
```

Step 4: Switch to the ORCLPDB Container and verify the connection name

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> alter session set container = orclpdb;

Session altered.

SQL> show con_name;

CON_NAME
-----
ORCLPDB
```

Step 5: Unlock the HR Account

```
SQL> alter user hr identified by thaxaena account unlock;

User altered.

SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A
```

Step 6: Disconnect from SYS User

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> disconnect
Disconnected from Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0
```

Step 7: Connect Using the HR User

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> disconnect
Disconnected from Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0
SQL> connect hr/thaxaena@orclpdb
Connected.
```

Step 8: Verify the Connected User

```
SQL> exec my_info;
Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

SQL> show user;
USER is "HR"
```

5. DISPLAYING BASIC INFORMATION IN ORACLE DATABASE AND FORMATTING OUTPUT IN SQL PLUS.

Step 1: View Username, Account Status, and Default Tablespace

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> select username, account_status, default_tablespace from dba_users;
```

USERNAME	ACCOUNT_STATUS	DEFAULT_TABLESPACE
SYS	OPEN	SYSTEM
SYSTEM	OPEN	SYSTEM
XS\$NULL	EXPIRED & LOCKED	SYSTEM
OJVMSYS	LOCKED	SYSTEM
LBACSYS	LOCKED	SYSTEM
OUTLN	LOCKED	SYSTEM
SYS\$UMF	LOCKED	SYSTEM
DBSNMP	LOCKED	SYSAUX
APPQOSSYS	LOCKED	SYSAUX
DBSFUSER	LOCKED	SYSAUX
GGSYS	LOCKED	SYSAUX

USERNAME	ACCOUNT_STATUS	DEFAULT_TABLESPACE
ANONYMOUS	EXPIRED & LOCKED	SYSAUX
CTXSYS	EXPIRED & LOCKED	SYSAUX
DVSYS	LOCKED	SYSAUX

Step 2: Display the Current User

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> show user;
USER is "SYS"
```

Step 3: Display Privileges Granted to a User

```
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> select grantee,privilege from dba_sys_privs where grantee='HR';

no rows selected
```

Step 4: Display All Users in the Database

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> select username from dba_users;
```

USERNAME
SYS
SYSTEM
XS\$NULL
OJVMSYS
LBACSYS

Formatting Output in SQLPlus

Step 5: Set Line Size

```
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> set linesize 130;
```

Step 6: Format Column Display

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> set linesize 130;
SQL> COLUMN USERNAME FORMAT A20;
SQL> COLUMN ACCOUNT_STATUS FORMAT A20;
SQL> COLUMN DEFAULT_TABLESPACE FORMAT A20;
```

Step 7: Display Formatted Output

```
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> SELECT USERNAME,
2         ACCOUNT_STATUS,
3         DEFAULT_TABLESPACE
4   FROM DBA_USERS;

USERNAME                                ACCOUNT_STATUS    DEFAULT_TABLESPACE
-----
SYS                                     OPEN              SYSTEM
SYSTEM                                 OPEN              SYSTEM
XS$NULL                               EXPIRED & LOCKED  SYSTEM
OJVMSYS                               LOCKED            SYSTEM
LBACSYS                               LOCKED            SYSTEM
OUTLN                                  LOCKED            SYSTEM
SYS$UMF                               LOCKED            SYSTEM
DBSNMP                                LOCKED            SYSAUX
```

Step 8: Limit the Number of Results

```
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> SELECT USERNAME,
2         ACCOUNT_STATUS
3   FROM DBA_USERS
4  WHERE ROWNUM <= 5;

USERNAME                                ACCOUNT_STATUS
-----
SYS                                     OPEN
AUDSYS                                 LOCKED
SYSTEM                                 OPEN
SYSBACKUP                             LOCKED
SYSDG                                  LOCKED
```

6. WORKING WITH TABLESPACE AND DATA FILE

(a) Displaying Tablespaces

Step 1: Display All Tablespaces Using DBA_TABLESPACES

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> SELECT TABLESPACE_NAME,
2          STATUS,
3          CONTENTS
4  FROM DBA_TABLESPACES;
```

TABLESPACE_NAME	STATUS	CONTENTS
SYSTEM	ONLINE	PERMANENT
SYSAUX	ONLINE	PERMANENT
UNDOTBS1	ONLINE	UNDO
TEMP	ONLINE	TEMPORARY
USERS	ONLINE	PERMANENT

Step 2: Login as HR User

```
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> connect hr/thaxaena@orclpdb
Connected.
```

Step 3: Display Tablespace Associated with Current User

```
SQL> exec my_info
Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

SQL> SELECT TABLESPACE_NAME,
2          STATUS
3  FROM USER_TABLESPACES;
```

TABLESPACE_NAME	STATUS
SYSTEM	ONLINE
SYSAUX	ONLINE
UNDOTBS1	ONLINE
TEMP	ONLINE
USERS	ONLINE

Step 4: Connect Back as SYSDBA

```
SQL> exec my_info
Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

SQL> connect / as sysdba
Connected.
```

Step 5: Display Real-Time Information Using V\$TABLESPACE

```
SQL> exec my_info
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> SELECT TS#,
2         NAME
3 FROM V$TABLESPACE;

TS# NAME
-----
1 SYSAUX
0 SYSTEM
2 UNDOTBS1
4 USERS
3 TEMP
0 SYSTEM
1 SYSAUX
2 UNDOTBS1
3 TEMP
0 SYSTEM
1 SYSAUX
2 UNDOTBS1
3 TEMP
5 USERS
```

Step 6: Display Tablespaces with Their Types

```
SQL> exec my_info
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> SELECT TABLESPACE_NAME,
2         CONTENTS
3 FROM DBA_TABLESPACES;

TABLESPACE_NAME      CONTENTS
-----
SYSTEM               PERMANENT
SYSAUX               PERMANENT
UNDOTBS1             UNDO
TEMP                 TEMPORARY
USERS                PERMANENT
```

(b) Create a Permanent Tablespace and Display the Newly Created Tablespace

Step 7: Connect as SYSDBA

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> connect / as sysdba;
Connected.
```

Step 8: Verify the Current Container

```
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> show con_name

CON_NAME
-----
CDB$ROOT
SQL> alter session set container = orclpdb;

Session altered.
```

Step 9: Create a Permanent Tablespace

```
SQL> exec my_info
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> CREATE TABLESPACE suresh_tbs1_23
 2 DATAFILE 'C:\Oracle19C\oradata\ORCL\orclpdb\suresh_df1_23.dbf'
 3 SIZE 20M
 4 AUTOEXTEND ON
 5 NEXT 5M
 6 MAXSIZE 100M;

Tablespace created.
```

Step 10: Display the Newly Created Tablespace

```
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> SELECT TABLESPACE_NAME,
 2 STATUS,
 3 CONTENTS
 4 FROM DBA_TABLESPACES
 5 WHERE TABLESPACE_NAME = 'SURESH_TBS1_23';

TABLESPACE_NAME          STATUS      CONTENTS
-----
SURESH_TBS1_23           ONLINE      PERMANENT
```

Step 11: Display the List of Tablespaces

```
SQL> exec my_info
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> SELECT TABLESPACE_NAME,
 2 CONTENTS
 3 FROM DBA_TABLESPACES
 4 ORDER BY TABLESPACE_NAME;

TABLESPACE_NAME          CONTENTS
-----
SURESH_TBS1_23           PERMANENT
SYSAUX                   PERMANENT
SYSTEM                   PERMANENT
TEMP                     TEMPORARY
UNDOTBS1                 UNDO
USERS                    PERMANENT
```

(c) Modify the Tablespace Created Above

Step 12: Change the Tablespace Status to OFFLINE

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> alter tablespace suresh_tbs1_23 offline;

Tablespace altered.
```

Step 13: Verify the Tablespace Status

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> SELECT TABLESPACE_NAME,
2         STATUS
3   FROM DBA_TABLESPACES
4  WHERE TABLESPACE_NAME = 'SURESH_TBS1_23';

TABLESPACE_NAME          STATUS
-----
SURESH_TBS1_23          OFFLINE
```

Step 14: Change the Tablespace Status to ONLINE

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> ALTER TABLESPACE suresh_tbs1_23 ONLINE;

Tablespace altered.
```

Step 15: Verify the Tablespace Status Again

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> SELECT TABLESPACE_NAME,
2         STATUS
3   FROM DBA_TABLESPACES
4  WHERE TABLESPACE_NAME = 'SURESH_TBS1_23';

TABLESPACE_NAME          STATUS
-----
SURESH_TBS1_23          ONLINE
```

Step 16: Add a New Data File to the Tablespace

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> ALTER TABLESPACE suresh_tbs1_23
2  ADD DATAFILE 'C:\Oracle19C\oradata\ORCL\orclpdb\suresh_df2_23.dbf'
3  SIZE 20M
4  AUTOEXTEND ON
5  NEXT 5M
6  MAXSIZE 100M;

Tablespace altered.
```

Step 17: Verify the Data Files

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> SELECT TABLESPACE_NAME,
2          FILE_NAME
3 FROM DBA_DATA_FILES
4 WHERE TABLESPACE_NAME = 'SURESH_TBS1_23';

TABLESPACE_NAME          FILE_NAME
-----
SURESH_TBS1_23           C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\SURESH_DF1_23.DB
SURESH_TBS1_23           C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\SURESH_DF2_23.DB
```

Step 18: Rename the Tablespace

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> ALTER TABLESPACE suresh_tbs1_23
2  RENAME TO suresh_dahal_tbs1_23;

Tablespace altered.
```

Step 19: Verify the Renamed Tablespace

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> SELECT TABLESPACE_NAME
2 FROM DBA_TABLESPACES
3 WHERE TABLESPACE_NAME = 'SURESH_DAHAL_TBS1_23';

TABLESPACE_NAME
-----
SURESH_DAHAL_TBS1_23
```

(d) Create and Delete a Tablespace

Step 20: Create a New Tablespace

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> CREATE TABLESPACE test_tbs_by_suresh
2 DATAFILE 'C:\Oracle19C\oradata\ORCL\orclpdb\test_df_by_suresh.dbf'
3 SIZE 20M
4 AUTOEXTEND ON
5 NEXT 5M
6 MAXSIZE 100M;

Tablespace created.
```

Step 21: Display the List of Tablespaces

```
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> SELECT TABLESPACE_NAME
       2 FROM DBA_TABLESPACES
       3 ORDER BY TABLESPACE_NAME;

TABLESPACE_NAME
-----
SURESH_DAHAL_TBS1_23
SYSAUX
SYSTEM
TEMP
TEST_TBS_BY_SURESH
UNDOTBS1
USERS
```

Step 22: Delete the Tablespace

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> DROP TABLESPACE test_tbs_by_suresh
       2 INCLUDING CONTENTS
       3 AND DATAFILES;

Tablespace dropped.
```

Step 23: Display the List of Tablespaces Again

```
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> SELECT TABLESPACE_NAME
       2 FROM DBA_TABLESPACES
       3 ORDER BY TABLESPACE_NAME;

TABLESPACE_NAME
-----
SURESH_DAHAL_TBS1_23
SYSAUX
SYSTEM
TEMP
UNDOTBS1
USERS
```

e) Displaying Data Files

Step 24: Display Information About Data Files Using DBA_DATA_FILES

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> SELECT FILE_NAME,
2         TABLESPACE_NAME,
3         BYTES,
4         AUTOEXTENSIBLE,
5         STATUS
6 FROM DBA_DATA_FILES;
```

FILE_NAME	TABLESPACE_NAME	BYTES	AUT	STATUS
C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\SYSTEM01.DBF	SYSTEM	283115520	YES	AVAILABLE
C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\SYS_AUX01.DBF	SYS_AUX	335544320	YES	AVAILABLE
C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\UNDOTBS01.DBF	UNDOTBS1	104857600	YES	AVAILABLE
C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\USERS01.DBF	USERS	5242880	YES	AVAILABLE
C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\SURESH_DF1_23.DB	SURESH_DAHAL_TBS1_23	20971520	YES	AVAILABLE
C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\SURESH_DF2_23.DB	SURESH_DAHAL_TBS1_23	20971520	YES	AVAILABLE

```
6 rows selected.
```

Step 25: Display Information About Data Files Using V\$DATAFILE

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> SELECT FILE#,
2         NAME,
3         STATUS
4 FROM V$DATAFILE;
```

FILE#	NAME	STATUS
9	C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\SYSTEM01.DBF	SYSTEM
10	C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\SYS_AUX01.DBF	ONLINE
11	C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\UNDOTBS01.DBF	ONLINE
12	C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\USERS01.DBF	ONLINE
13	C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\SURESH_DF1_23.DBF	ONLINE
14	C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\SURESH_DF2_23.DBF	ONLINE

```
6 rows selected.
```

f) Resize the Data File Created in (b)

Step 26: Resize the Data File

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> ALTER DATABASE
2 DATAFILE 'C:\Oracle19C\oradata\ORCL\orclpdb\suresh_df1_23.dbf'
3 RESIZE 50M;
```

```
Database altered.
```

Step 27: Verify the Resized Data File

```
SQL> exec my_info;
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.
```

```
SQL> SELECT FILE_NAME,
2         BYTES/1024/1024 AS SIZE_MB
3   FROM DBA_DATA_FILES
4  WHERE FILE_NAME LIKE '%SURESH_DF1_23.DBF';
```

FILE_NAME	SIZE_MB
C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\SURESH_DF1_23.DB	50

7. WORKING WITH ORACLE SQL DEVELOPER

Step 1: Install Oracle SQL Developer



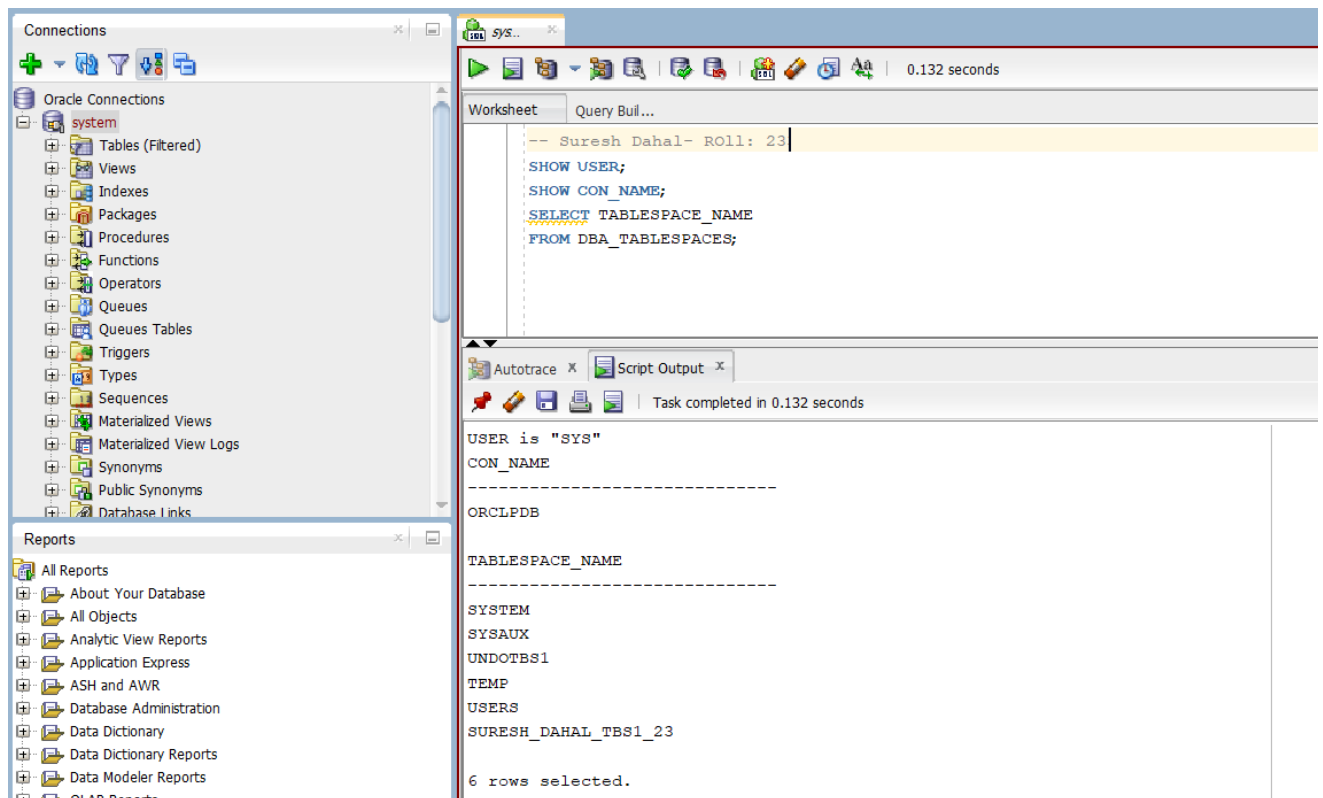
Part A: Create a Connection for SYSTEM User

Step 2: Create a New Database Connection

New / Select Database Connection

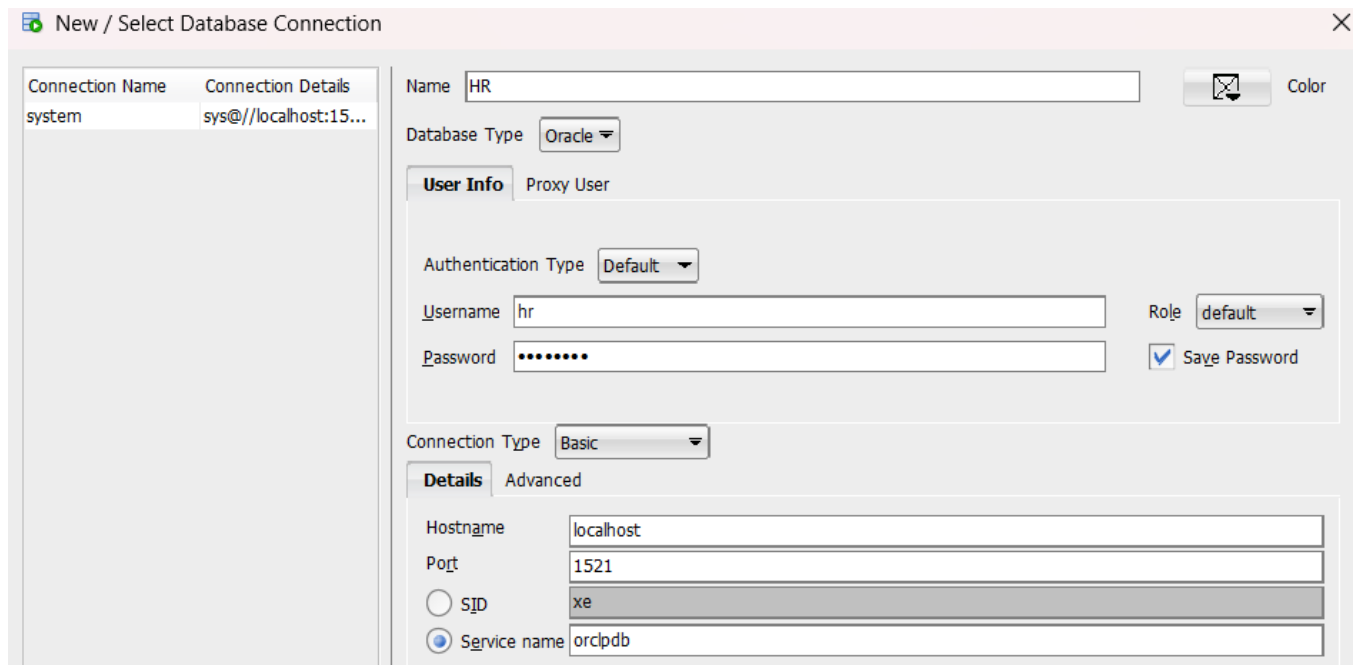
Connection Name	Connection Details
	Name: system Database Type: Oracle User Info Proxy User Authentication Type: Default Username: sys Role: SYSDBA Password: ***** ☒ Save Password Connection Type: Basic Details Advanced Hostname: localhost Port: 1521 ☐ SID xe ☒ Service name orclpdb

Step 3: Execute Commands from SQL Worksheet



Part B: Create a Connection for HR User

Step 4: Create a New Database Connection



Step 5: Execute Commands from SQL Worksheet

The screenshot displays the SQL Developer interface with the following components:

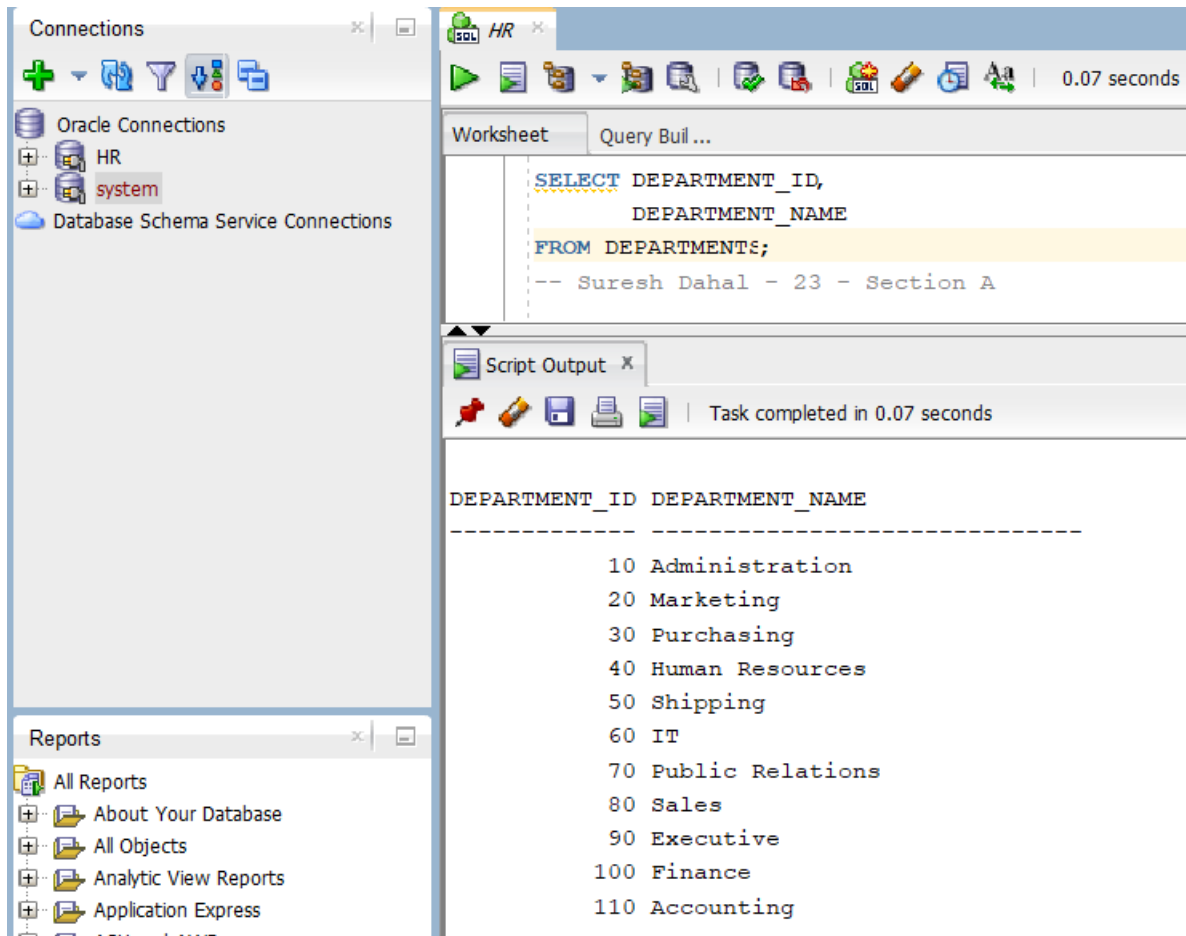
- Connections:** A tree view on the left showing 'Oracle Connections' with 'HR' and 'system' schemas, and 'Database Schema Service Connections'.
- Worksheet:** The main area for writing SQL. It shows the following commands:

```
SHOW USER;  
SHOW CON_NAME;  
SELECT EMPLOYEE_ID,  
       FIRST_NAME,  
       LAST_NAME  
FROM EMPLOYEES;  
-- Suresh Dahal - 23 - Section A
```
- Script Output:** A panel below the worksheet showing the results of the executed commands:

```
USER is "HR"  
CON_NAME  
-----  
ORCLPDB
```
- Reports:** A tree view on the bottom left showing various report categories like 'All Reports', 'About Your Database', 'All Objects', etc.
- Table:** The results of the SELECT query are displayed in a table format:

EMPLOYEE_ID	FIRST_NAME	LAST_NAME
100	Steven	King
101	Neena	Kochhar
102	Lex	De Haan
103	Alexander	Hunold
104	Bruce	Ernst
105	David	Austin
106	Valli	Pataballa
107	Diana	Lorentz
108	Nancy	Greenberg
109	Daniel	Faviet
110	John	Chen

Step 6: Execute Any Additional Query



The screenshot displays the SQL Developer interface with the following components:

- Connections:** Shows Oracle Connections with 'HR' and 'system' selected, and Database Schema Service Connections.
- Worksheet:** Contains the SQL query:

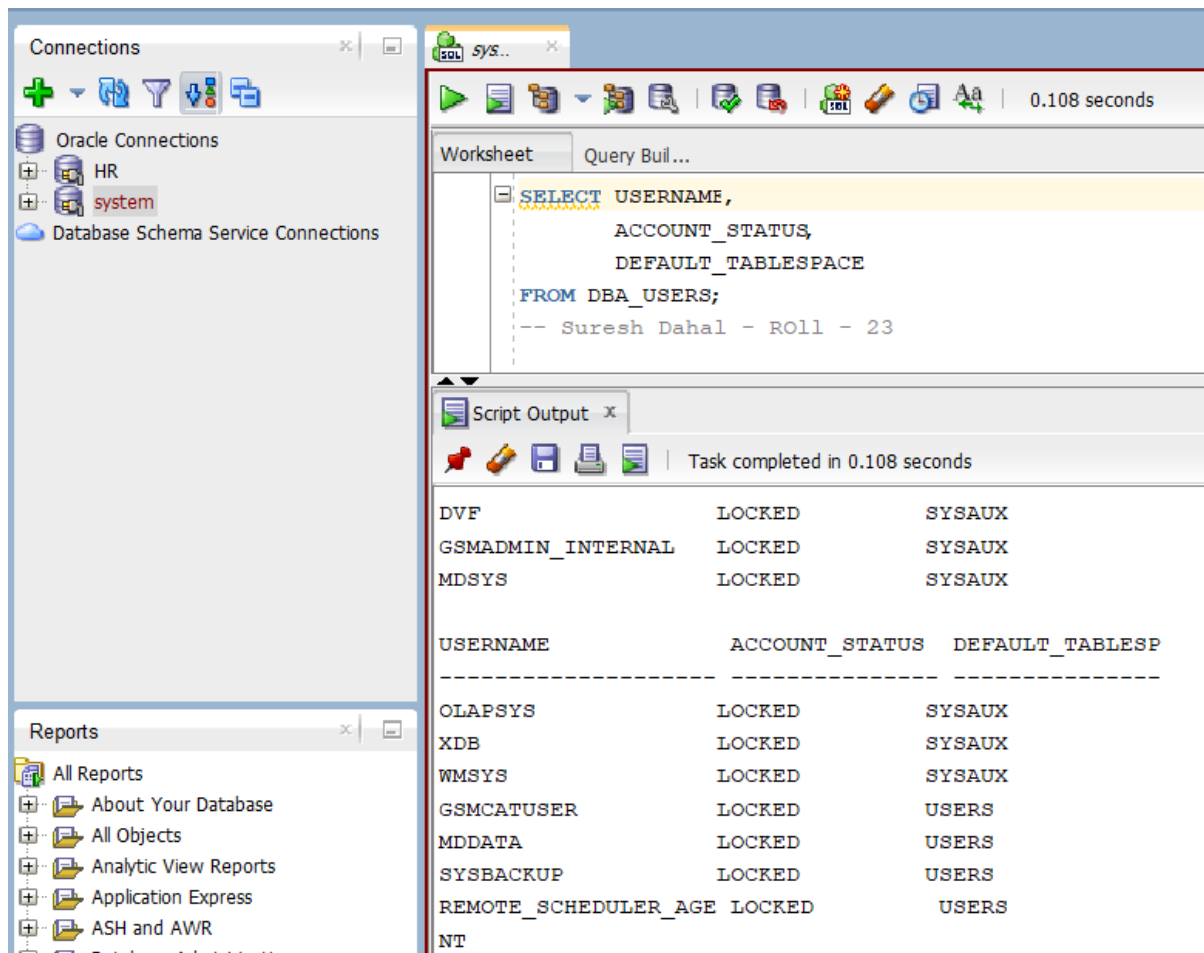
```
SELECT DEPARTMENT_ID,  
       DEPARTMENT_NAME  
FROM DEPARTMENTS;  
-- Suresh Dahal - 23 - Section A
```
- Script Output:** Shows the execution result:

```
DEPARTMENT_ID DEPARTMENT_NAME  
-----  
10 Administration  
20 Marketing  
30 Purchasing  
40 Human Resources  
50 Shipping  
60 IT  
70 Public Relations  
80 Sales  
90 Executive  
100 Finance  
110 Accounting
```

8. USER MANAGEMENT WITH GRANT AND REVOKE PRIVILEGES.

(a) Display Information About Users Using DBA_USERS

Step 1: Display Information About Users



The screenshot shows the Oracle SQL Developer interface. On the left, the 'Connections' pane shows 'HR' and 'system' connections. The 'Reports' pane is also visible. The main window displays a query in the 'Query Builder' tab:

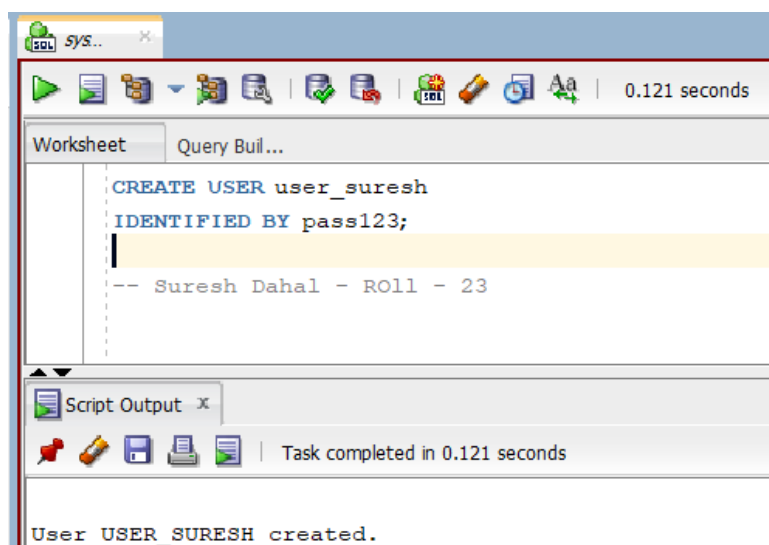
```
SELECT USERNAME,  
        ACCOUNT_STATUS,  
        DEFAULT_TABLESPACE  
FROM DBA_USERS;  
-- Suresh Dahal - Roll - 23
```

The 'Script Output' pane shows the results of the query, indicating 'Task completed in 0.108 seconds'. The output is a table with three columns: USERNAME, ACCOUNT_STATUS, and DEFAULT_TABLESPACE.

USERNAME	ACCOUNT_STATUS	DEFAULT_TABLESPACE
DVF	LOCKED	SYSAUX
GSMADMIN_INTERNAL	LOCKED	SYSAUX
MDSYS	LOCKED	SYSAUX
OLAPSYS	LOCKED	SYSAUX
XDB	LOCKED	SYSAUX
WMSYS	LOCKED	SYSAUX
GSMCATUSER	LOCKED	USERS
MDDATA	LOCKED	USERS
SYSDBACKUP	LOCKED	USERS
REMOTE_SCHEDULER_AGENT	LOCKED	USERS
NT		

b) Create a User

Step 2: Create User user_suresh



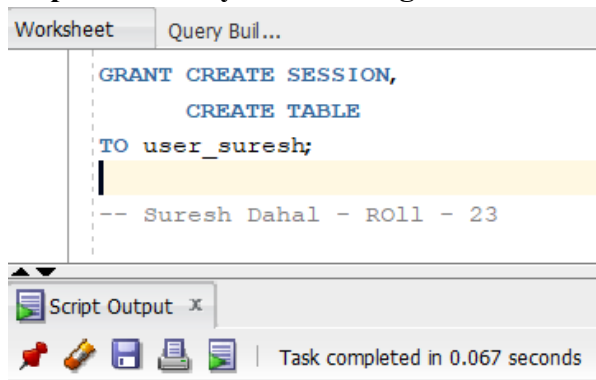
The screenshot shows the Oracle SQL Developer interface. The 'Query Builder' tab contains the following SQL command:

```
CREATE USER user_suresh  
IDENTIFIED BY pass123;  
-- Suresh Dahal - Roll - 23
```

The 'Script Output' pane shows the results of the command, indicating 'Task completed in 0.121 seconds'. The output is a single line: 'User USER_SURESH created.'

(c) Grant Privileges to the User

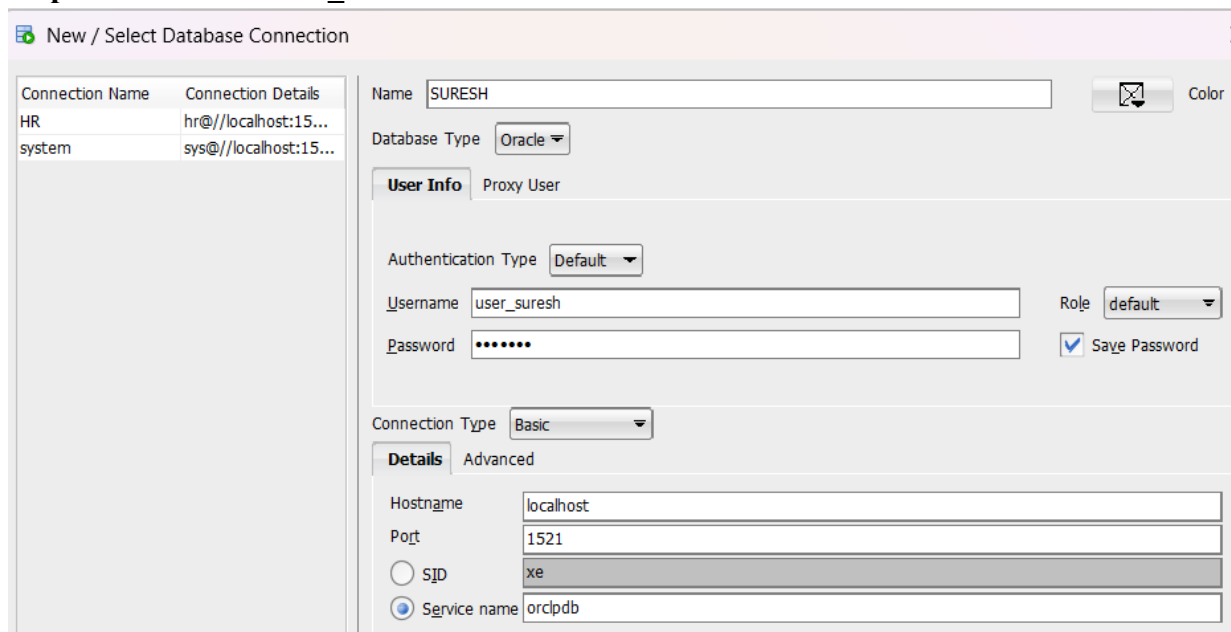
Step 3: Grant System Privileges



Grant succeeded.

(d) Create a Table After Logging in as user_suresh

Step 4: Connect as user_suresh



Step 5: Create a Table

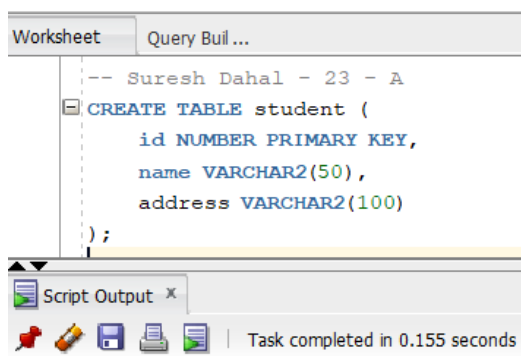
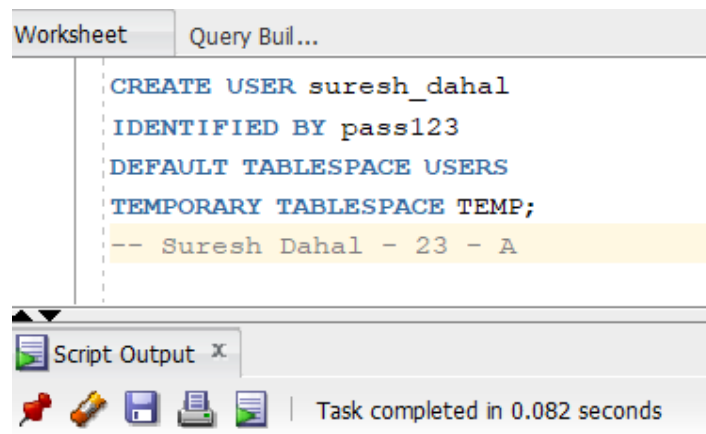


Table STUDENT created.

(e) Create Another User with Tablespace Clause

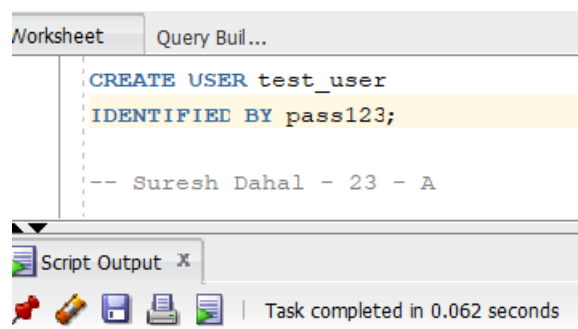
Step 6: Create User suresh_dahal



User SURESH_DAHAL created.

(f) Create Another User of Your Choice

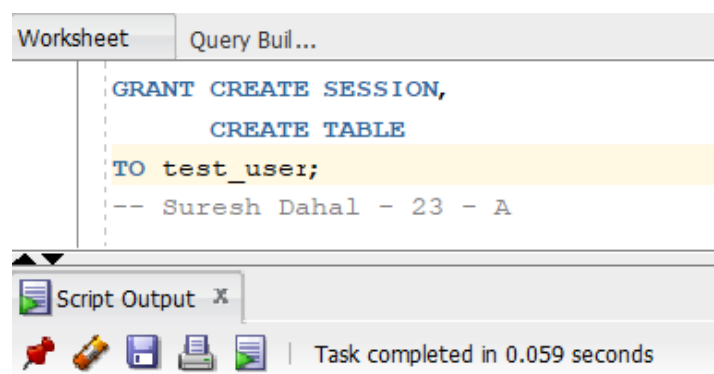
Step 7: Create User



User TEST_USER created.

(g) Grant Privileges to the User Created in (f)

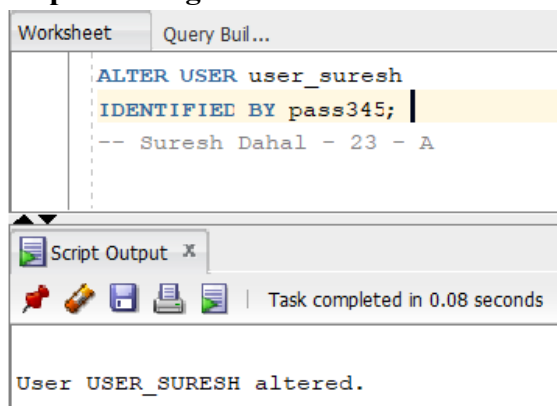
Step 8: Grant Privileges



Grant succeeded.

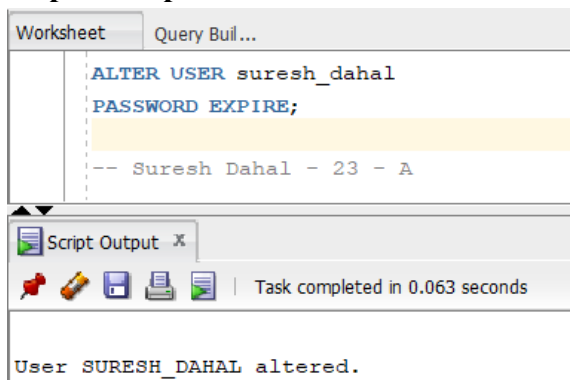
(h) Change the Password of user_suresh

Step 9: Change Password



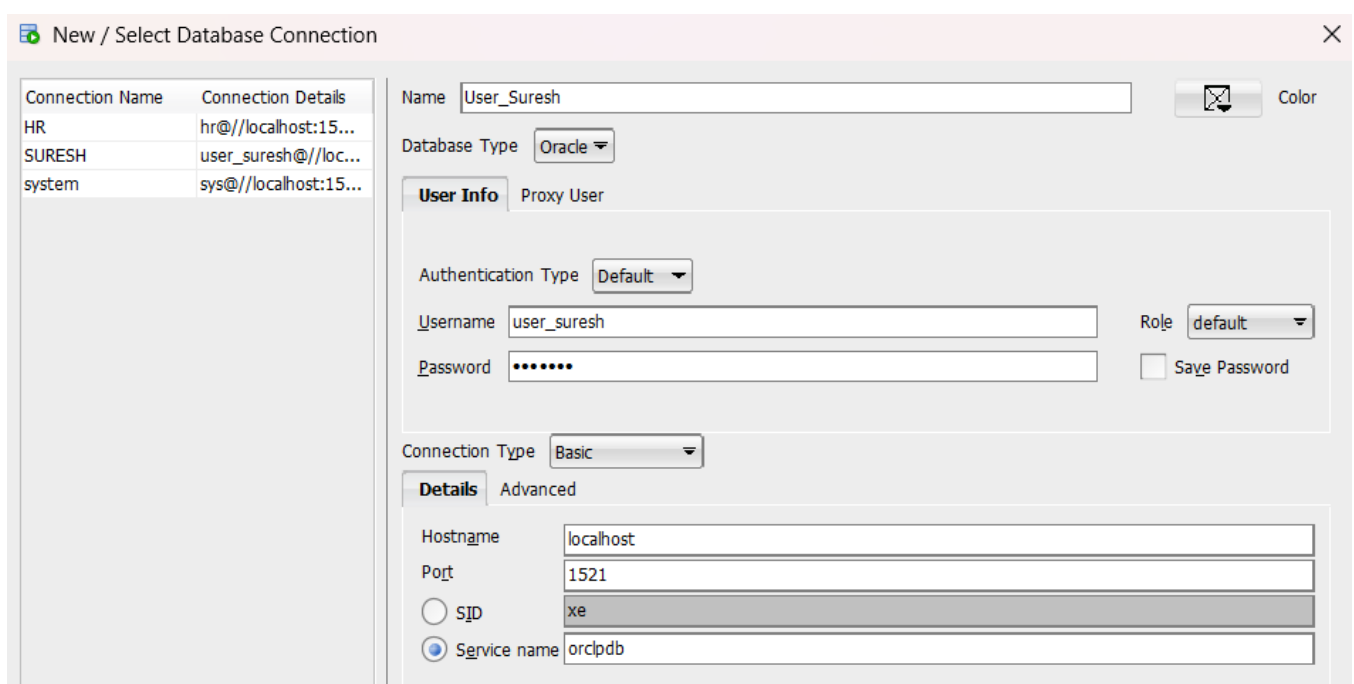
(i) Enable Password Expiration for suresh_dahal

Step 10: Expire Password

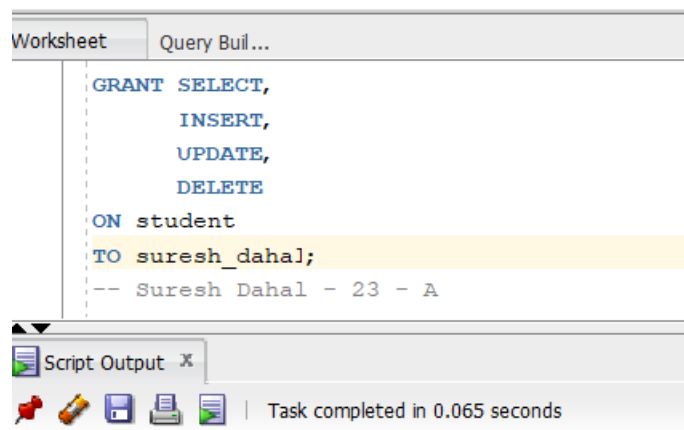


(j) Grant Object Privileges on the Table

Step 11: Connect as user_suresh



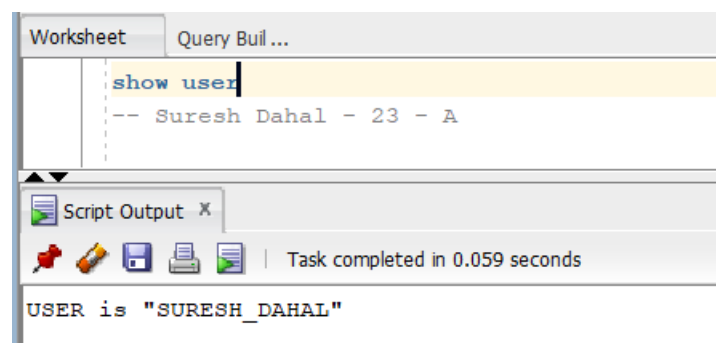
Step 12: Grant Object Privileges



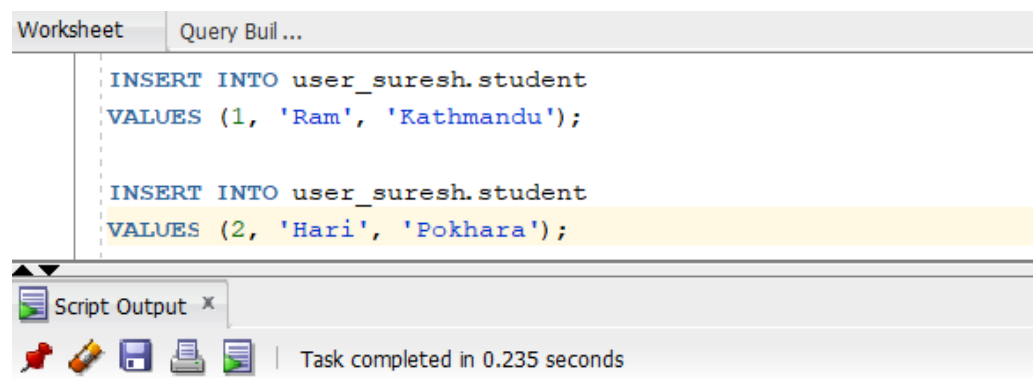
Grant succeeded.

(k) Insert, Update and Select Records by Logging in as suresh_dahal

Step 13: Connect as suresh_dahal



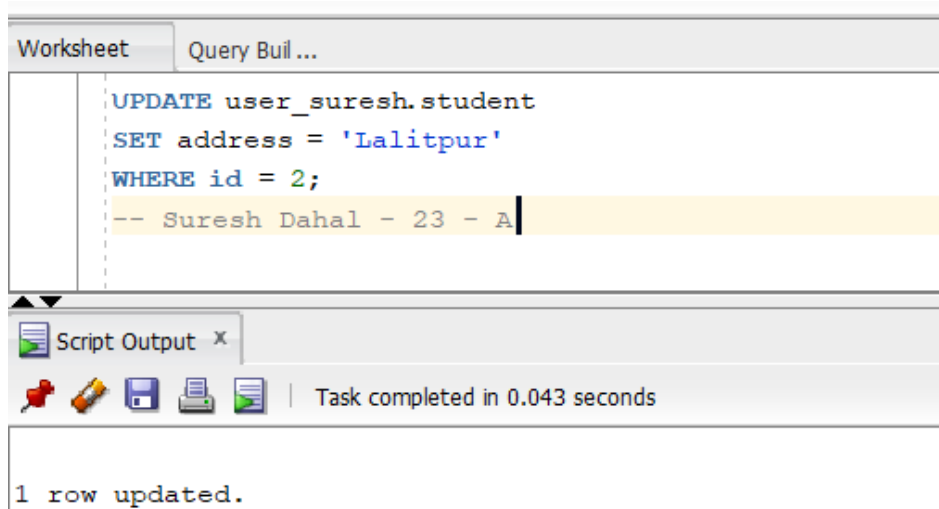
Step 14: Insert Two Records



1 row inserted.

1 row inserted.

Step 15: Update the Second Record

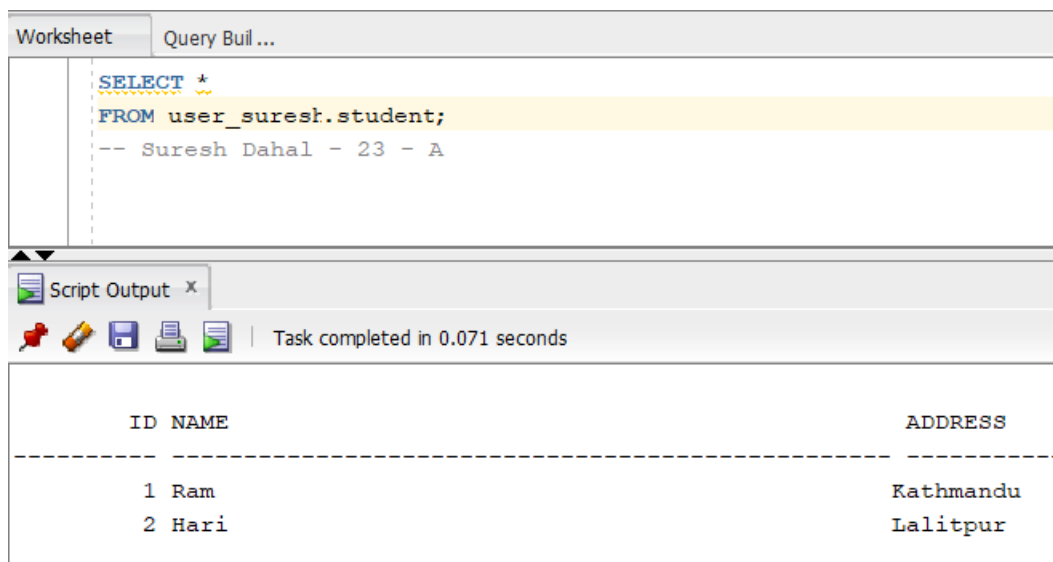


The screenshot shows a database query editor with a 'Worksheet' tab and a 'Query Build...' button. The query text is:

```
UPDATE user_suresh.student
SET address = 'Lalitpur'
WHERE id = 2;
-- Suresh Dahal - 23 - A
```

Below the query editor, the 'Script Output' tab shows the result: '1 row updated.' The task completed in 0.043 seconds.

Step 16: Display All Records



The screenshot shows a database query editor with a 'Worksheet' tab and a 'Query Build...' button. The query text is:

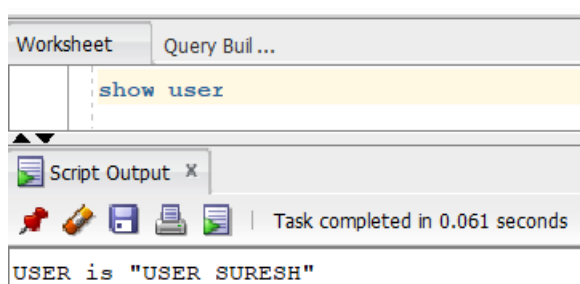
```
SELECT *
FROM user_suresh.student;
-- Suresh Dahal - 23 - A
```

Below the query editor, the 'Script Output' tab shows the result: 'Task completed in 0.071 seconds'. The output is a table with three columns: ID, NAME, and ADDRESS.

ID	NAME	ADDRESS
1	Ram	Kathmandu
2	Hari	Lalitpur

(I) Revoke DELETE Privilege

Step 17: Connect as user_suresh

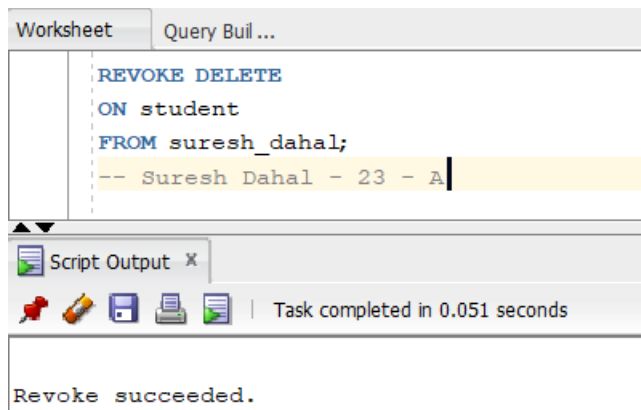


The screenshot shows a database query editor with a 'Worksheet' tab and a 'Query Build...' button. The query text is:

```
show user
```

Below the query editor, the 'Script Output' tab shows the result: 'Task completed in 0.061 seconds'. The output is: 'USER is "USER_SURESH"'

Step 18: Revoke DELETE Privilege



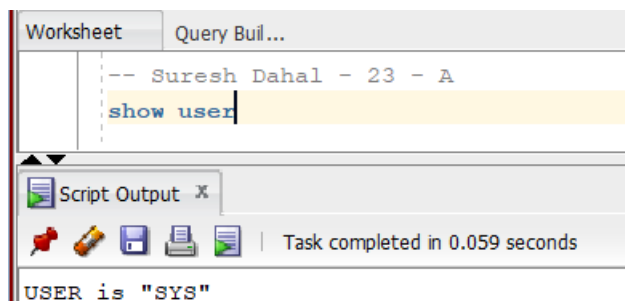
The screenshot shows the SQL Developer interface. The 'Worksheet' tab is active, displaying the following SQL code:

```
REVOKE DELETE
ON student
FROM suresh_dahal;
-- Suresh Dahal - 23 - A
```

Below the worksheet, the 'Script Output' window is open, showing the message: 'Task completed in 0.051 seconds' and 'Revoke succeeded.'

(m) Remove the User Created in (f)

Step 19: Connect as SYSDBA

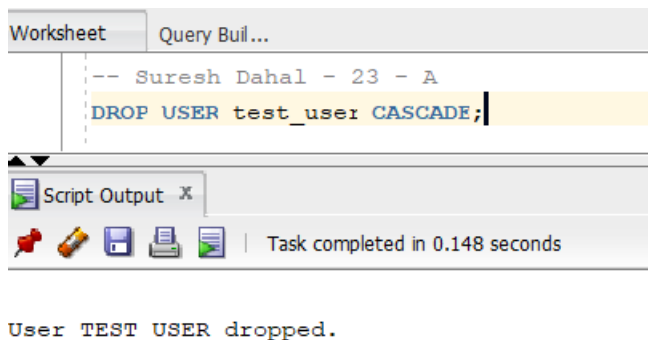


The screenshot shows the SQL Developer interface. The 'Worksheet' tab is active, displaying the following SQL code:

```
-- Suresh Dahal - 23 - A
show user
```

Below the worksheet, the 'Script Output' window is open, showing the message: 'Task completed in 0.059 seconds' and 'USER is "SYS"'

Step 20: Drop the User



The screenshot shows the SQL Developer interface. The 'Worksheet' tab is active, displaying the following SQL code:

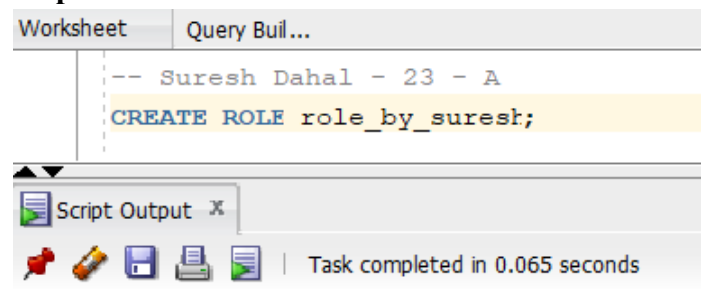
```
-- Suresh Dahal - 23 - A
DROP USER test_user CASCADE;
```

Below the worksheet, the 'Script Output' window is open, showing the message: 'Task completed in 0.148 seconds' and 'User TEST_USER dropped.'

9. WORKING WITH ROLES AND PRIVILEGES.

(a) Create a Role Named role_by_yourName

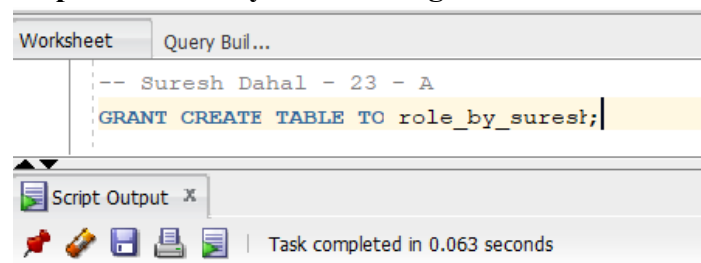
Step 1: Create the Role



Role ROLE_BY_SURESH created.

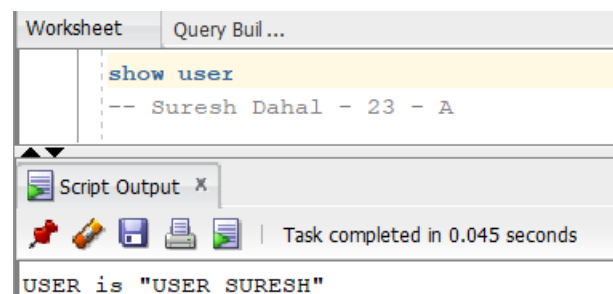
(b) Grant System and Object Privileges to the Role

Step 2: Grant a System Privilege

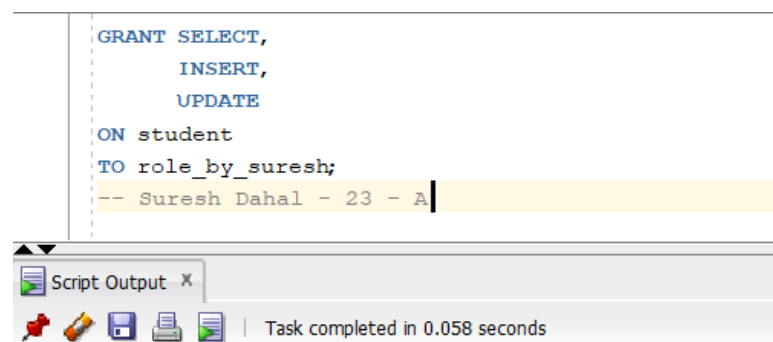


Grant succeeded.

Step 3: Connect as the Owner of the Table



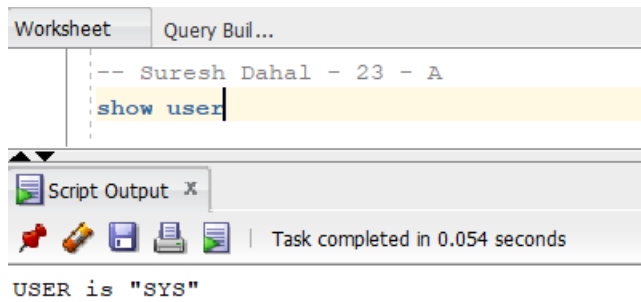
Step 4: Grant Object Privileges



Grant succeeded.

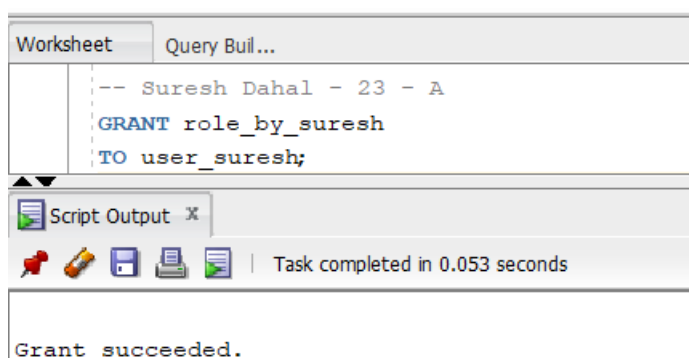
(c) Assign the Role to the User Created in Lab 8(b)

Step 5: Connect as SYSTEM



The screenshot shows the SQL Developer interface. The 'Worksheet' tab is active, displaying a query: `-- Suresh Dahal - 23 - A` followed by `show user` on a highlighted yellow background. Below the query, the 'Script Output' window shows the result: `USER is "SYS"`. The status bar indicates 'Task completed in 0.054 seconds'.

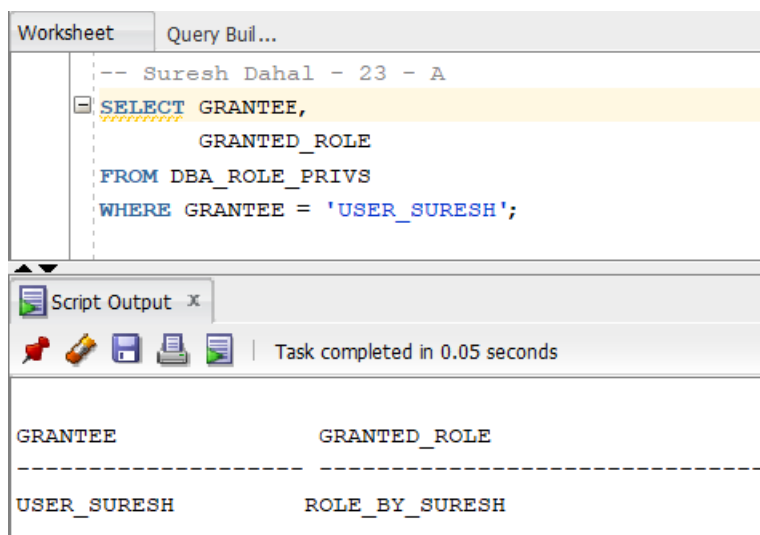
Step 6: Grant the Role



The screenshot shows the SQL Developer interface. The 'Worksheet' tab is active, displaying a query: `-- Suresh Dahal - 23 - A` followed by `GRANT role_by_suresh` and `TO user_suresh;` on a highlighted yellow background. Below the query, the 'Script Output' window shows the result: `Grant succeeded.`. The status bar indicates 'Task completed in 0.053 seconds'.

(d) View Roles Granted to a User

Step 7: Display Roles Granted to the User



The screenshot shows the SQL Developer interface. The 'Worksheet' tab is active, displaying a query: `-- Suresh Dahal - 23 - A` followed by `SELECT GRANTEE,` (highlighted on a yellow background), `GRANTED_ROLE`, `FROM DBA_ROLE_PRIVS`, and `WHERE GRANTEE = 'USER_SURESH';`. Below the query, the 'Script Output' window shows the result in a table format. The status bar indicates 'Task completed in 0.05 seconds'.

GRANTEE	GRANTED_ROLE
USER_SURESH	ROLE_BY_SURESH

(e) Display Privileges Granted to the Role

Step 8: Display System Privileges

The screenshot shows the SQL Developer interface with the Query Builder tab active. The query editor contains the following SQL statement:

```
-- Suresh Dahal - 23 - A  
SELECT ROLE,  
       PRIVILEGE  
FROM ROLE_SYS_PRIVS  
WHERE ROLE = 'ROLE_BY_SURESH';
```

Below the query editor, the Script Output window shows the execution status: "Task completed in 0.072 seconds".

ROLE	PRIVILEGE

ROLE_BY_SURESH	CREATE TABLE

Step 9: Display Object Privileges

The screenshot shows the SQL Developer interface with the Query Builder tab active. The query editor contains the following SQL statement:

```
-- Suresh Dahal - 23 - A  
SELECT GRANTEE,  
       TABLE_NAME,  
       PRIVILEGE  
FROM DBA_TAB_PRIVS  
WHERE GRANTEE = 'ROLE_BY_SURESH';
```

Below the query editor, the Script Output window shows the execution status: "Task completed in 0.058 seconds".

GRANTEE	TABLE_NAME	PRIVILEGE

ROLE_BY_SURESH	STUDENT	INSERT
ROLE_BY_SURESH	STUDENT	SELECT
ROLE_BY_SURESH	STUDENT	UPDATE

(f) Create Another Role

Step 10: Create a New Role

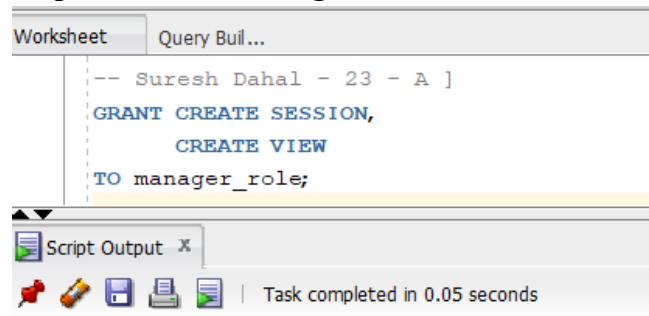
The screenshot shows the SQL Developer interface with the Query Builder tab active. The query editor contains the following SQL statement:

```
-- Suresh Dahal - 23 - A ]  
create role manager_role
```

Below the query editor, the Script Output window shows the execution status: "Task completed in 0.045 seconds".

Role MANAGER_ROLE created.

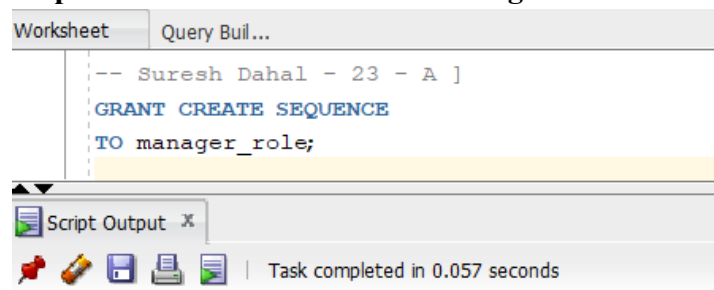
Step 11: Grant Privileges



Grant succeeded.

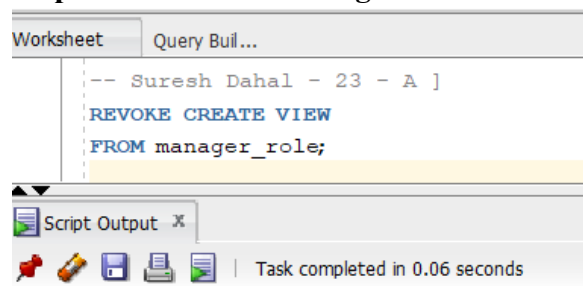
(g) Modify the Role Created in (f)

Step 12: Grant an Additional Privilege



Grant succeeded.

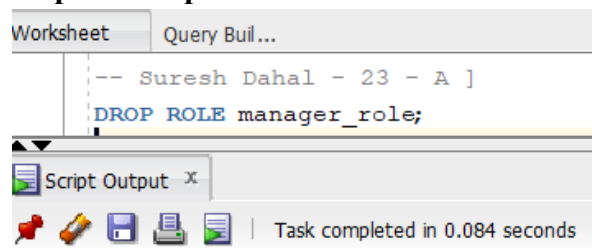
Step 13: Revoke a Privilege



Revoke succeeded.

(h) Drop the Role Created in (f)

Step 14: Drop the Role

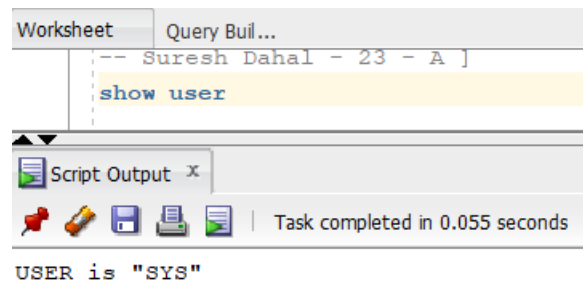


Role MANAGER_ROLE dropped.

10. WORKING WITH PROFILES.

(a) Create a Profile Named profile_by_yourName

Step 1: Connect as SYSTEM



Worksheet Query Buil...

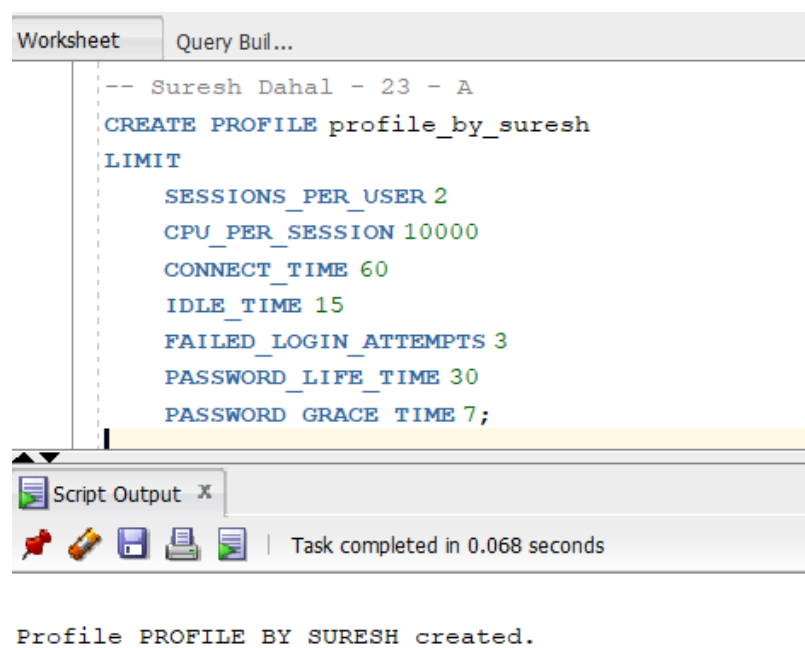
```
-- Suresh Dahal - 23 - A ]  
show user
```

Script Output x

Task completed in 0.055 seconds

USER is "SYS"

Step 2: Create the Profile



Worksheet Query Buil...

```
-- Suresh Dahal - 23 - A  
CREATE PROFILE profile_by_suresh  
LIMIT  
    SESSIONS_PER_USER 2  
    CPU_PER_SESSION 10000  
    CONNECT_TIME 60  
    IDLE_TIME 15  
    FAILED_LOGIN_ATTEMPTS 3  
    PASSWORD_LIFE_TIME 30  
    PASSWORD_GRACE_TIME 7;
```

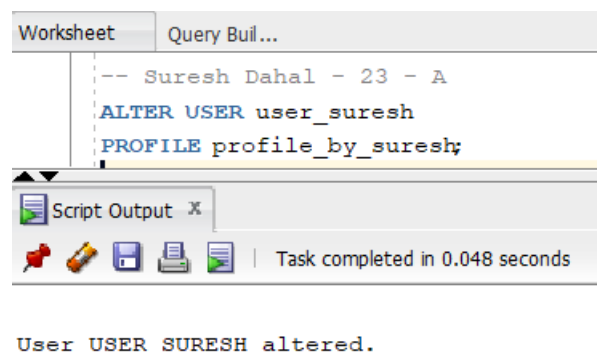
Script Output x

Task completed in 0.068 seconds

Profile PROFILE_BY_SURESH created.

(b) Assign the Profile to the User Created in Practical 8(b)

Step 3: Assign the Profile



Worksheet Query Buil...

```
-- Suresh Dahal - 23 - A  
ALTER USER user_suresh  
PROFILE profile_by_suresh;
```

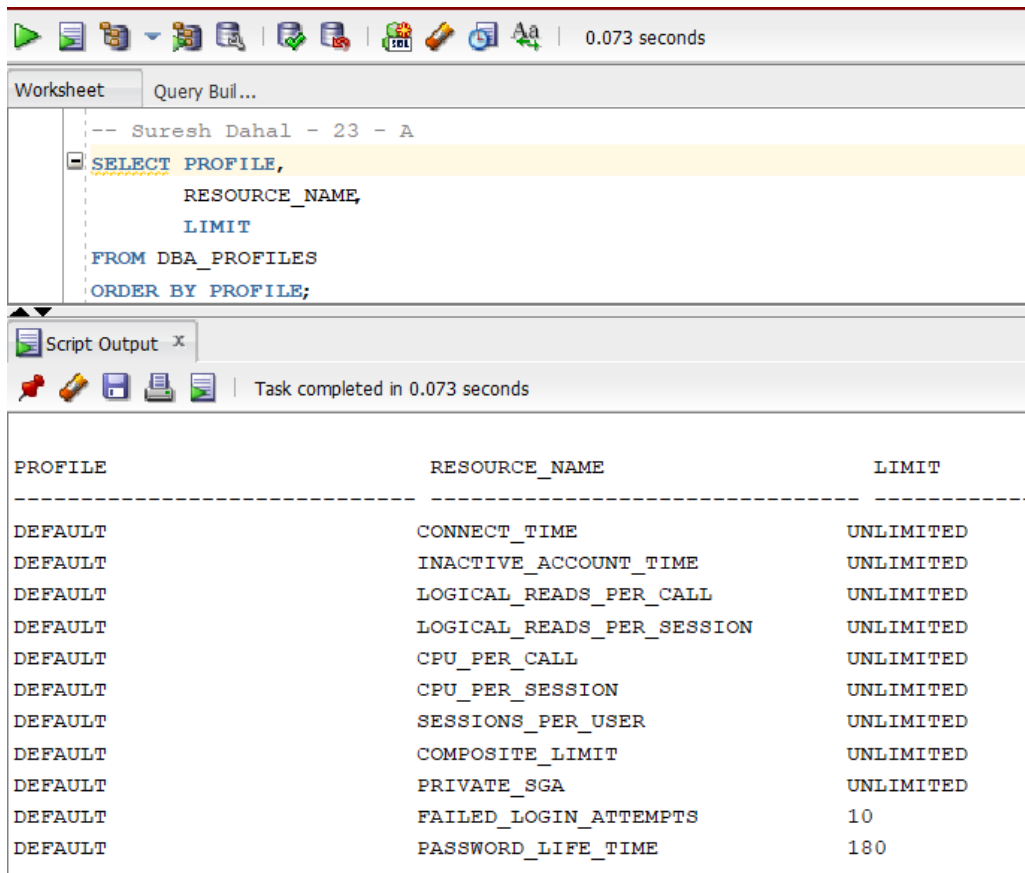
Script Output x

Task completed in 0.048 seconds

User USER_SURESH altered.

(c) View Profile Information

Step 4: Display All Profiles



The screenshot shows the SQL Developer interface. The top toolbar includes icons for running queries, saving, and other database functions. The status bar at the top right indicates "0.073 seconds". The "Worksheet" tab is active, displaying a SQL query:

```
-- Suresh Dahal - 23 - A
SELECT PROFILE,
       RESOURCE_NAME,
       LIMIT
FROM DBA_PROFILES
ORDER BY PROFILE;
```

The "Script Output" tab is also visible, showing the results of the query. The output is a table with three columns: PROFILE, RESOURCE_NAME, and LIMIT. The data is as follows:

PROFILE	RESOURCE_NAME	LIMIT
DEFAULT	CONNECT_TIME	UNLIMITED
DEFAULT	INACTIVE_ACCOUNT_TIME	UNLIMITED
DEFAULT	LOGICAL_READS_PER_CALL	UNLIMITED
DEFAULT	LOGICAL_READS_PER_SESSION	UNLIMITED
DEFAULT	CPU_PER_CALL	UNLIMITED
DEFAULT	CPU_PER_SESSION	UNLIMITED
DEFAULT	SESSIONS_PER_USER	UNLIMITED
DEFAULT	COMPOSITE_LIMIT	UNLIMITED
DEFAULT	PRIVATE_SGA	UNLIMITED
DEFAULT	FAILED_LOGIN_ATTEMPTS	10
DEFAULT	PASSWORD_LIFE_TIME	180

Step 5: Display Information About the Created Profile

PROFILE	RESOURCE_NAME	LIMIT
PROFILE_BY_SURESH	COMPOSITE_LIMIT	DEFAULT
PROFILE_BY_SURESH	SESSIONS_PER_USER	2
PROFILE_BY_SURESH	CPU_PER_SESSION	10000
PROFILE_BY_SURESH	CPU_PER_CALL	DEFAULT
PROFILE_BY_SURESH	LOGICAL_READS_PER_SESSION	DEFAULT
PROFILE_BY_SURESH	LOGICAL_READS_PER_CALL	DEFAULT
PROFILE_BY_SURESH	IDLE_TIME	15
PROFILE_BY_SURESH	CONNECT_TIME	60
PROFILE_BY_SURESH	PRIVATE_SGA	DEFAULT
PROFILE_BY_SURESH	FAILED_LOGIN_ATTEMPTS	3
PROFILE_BY_SURESH	PASSWORD_LIFE_TIME	30

PROFILE	RESOURCE_NAME	LIMIT
PROFILE_BY_SURESH	PASSWORD_REUSE_TIME	DEFAULT
PROFILE_BY_SURESH	PASSWORD_REUSE_MAX	DEFAULT
PROFILE_BY_SURESH	PASSWORD_VERIFY_FUNCTION	DEFAULT
PROFILE_BY_SURESH	PASSWORD_LOCK_TIME	DEFAULT

(d) Create Another Profile

Step 6: Create a New Profile

```
Worksheet  Query Bul...
-- Suresh Dahal - 23 - A
CREATE PROFILE test_profile
LIMIT
  SESSIONS_PER_USER 1
  IDLE_TIME 10
  PASSWORD LIFE TIME 20;
```

Script Output x

Task completed in 0.065 seconds

Profile TEST_PROFILE created.

(e) Modify the Profile and Display Updated Information

Step 7: Modify the Profile

```
Worksheet  Query Bul...
-- Suresh Dahal - 23 - A
ALTER PROFILE test_profile
LIMIT
  IDLE_TIME 20
  PASSWORD LIFE TIME 45;
```

Script Output x

Task completed in 0.071 seconds

Profile TEST_PROFILE altered.

Step 8: Display the Updated Profile

```
-- Suresh Dahal - 23 - A
SELECT PROFILE,
  RESOURCE_NAME,
  LIMIT
FROM DBA_PROFILES
WHERE PROFILE = 'TEST PROFILE';
```

Script Output x

Task completed in 0.081 seconds

PROFILE	RESOURCE_NAME	LIMIT
TEST_PROFILE	COMPOSITE_LIMIT	DEFAULT
TEST_PROFILE	SESSIONS_PER_USER	1
TEST_PROFILE	CPU_PER_SESSION	DEFAULT
TEST_PROFILE	CPU_PER_CALL	DEFAULT
TEST_PROFILE	LOGICAL_READS_PER_SESSION	DEFAULT
TEST_PROFILE	LOGICAL_READS_PER_CALL	DEFAULT
TEST_PROFILE	IDLE_TIME	20
TEST_PROFILE	CONNECT_TIME	DEFAULT
TEST_PROFILE	PRIVATE_SGA	DEFAULT
TEST_PROFILE	FAILED_LOGIN_ATTEMPTS	DEFAULT

(f) Drop the Profile and Display Updated Information

Step 9: Drop the Profile

The screenshot shows the SQL Developer interface. The 'Worksheet' tab is active, displaying a SQL script: `-- Suresh Dahal - 23 - A` followed by `DROP PROFILE test_profile CASCADE;`. The 'Script Output' tab is also visible, showing the message 'Task completed in 0.07 seconds' and 'Profile TEST_PROFILE dropped.'

Step 10: Display the Remaining Profiles

The screenshot shows the SQL Developer interface. The 'Worksheet' tab is active, displaying a SQL script: `-- Suresh Dahal - 23 - A` followed by `SELECT PROFILE,`
`RESOURCE_NAME,`
`LIMIT`
`FROM DBA_PROFILES`
`ORDER BY PROFILE;`. The 'Script Output' tab is also visible, showing the message 'Task completed in 0.087 seconds'. Below the script output, a table displays the results of the query.

PROFILE	RESOURCE_NAME	LIMIT
ORA_STIG_PROFILE	FAILED_LOGIN_ATTEMPTS	3
PROFILE_BY_SURESH	COMPOSITE_LIMIT	DEFAULT
PROFILE_BY_SURESH	INACTIVE_ACCOUNT_TIME	DEFAULT
PROFILE_BY_SURESH	CPU_PER_SESSION	10000
PROFILE_BY_SURESH	CPU_PER_CALL	DEFAULT
PROFILE_BY_SURESH	LOGICAL_READS_PER_SESSION	DEFAULT
PROFILE_BY_SURESH	LOGICAL_READS_PER_CALL	DEFAULT
PROFILE_BY_SURESH	IDLE_TIME	15
PROFILE_BY_SURESH	CONNECT_TIME	60
PROFILE_BY_SURESH	PASSWORD_GRACE_TIME	7
PROFILE_BY_SURESH	PASSWORD_LOCK_TIME	DEFAULT